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Wu et al.

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(54) **ELECTRONIC DEVICE**

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(51) **Int. Cl.**

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G06F 3/041	(2006.01)
G06F 3/046	(2006.01)
H05K 1/02	(2006.01)

(52) **U.S. Cl.**

CPC ***H05K 1/0277*** (2013.01); ***G06F 3/03545*** (2013.01); ***G06F 3/0412*** (2013.01); ***G06F 3/046*** (2013.01); *G06F 2203/04112* (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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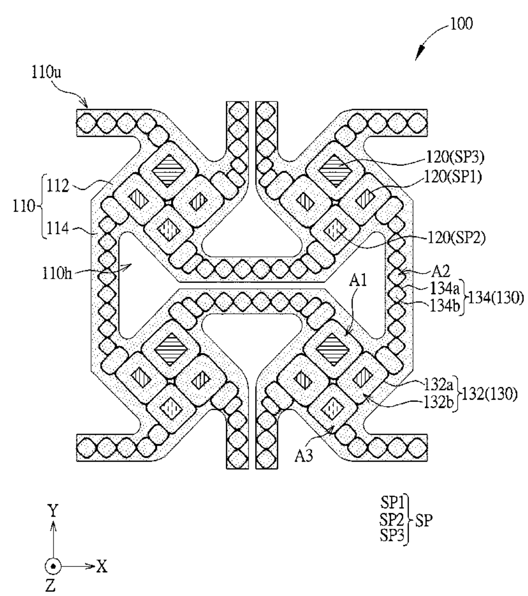
(57) **ABSTRACT**

An electronic device includes a stretchable substrate, a plurality of first electronic units and a plurality of second electronic units. The stretchable substrate has a plurality of island portions and a plurality of bridge portions, and each of the bridge portions connecting at least two of the island portions. The first electronic units are disposed on the island portions. The second electronic units are disposed on the stretchable substrate, each of the second electronic units includes a mesh pattern disposed on at least one of the island portions, and the mesh pattern includes a first opening overlapped with at least one of the first electronic units.

15 Claims, 13 Drawing Sheets



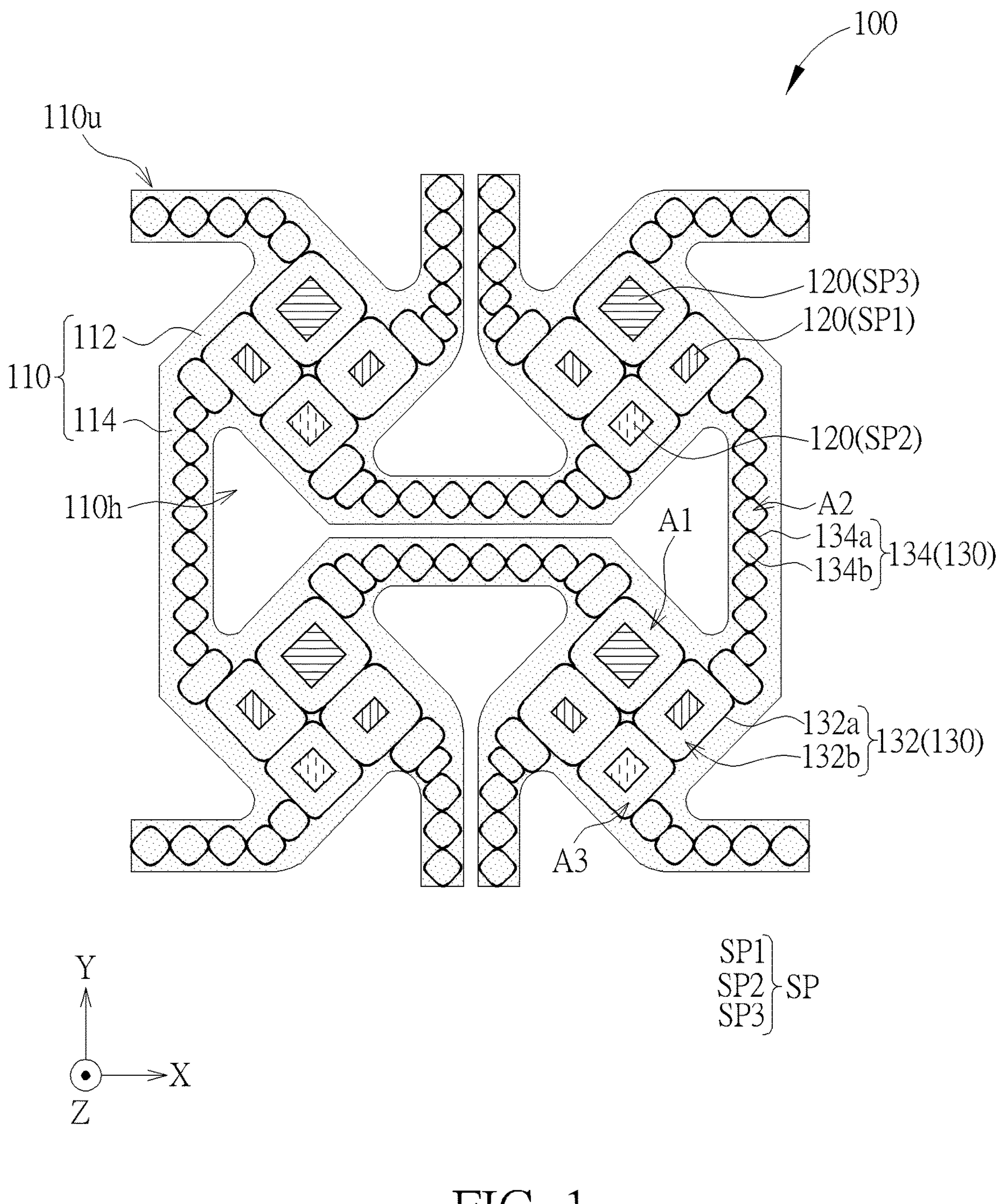

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FIG. 1

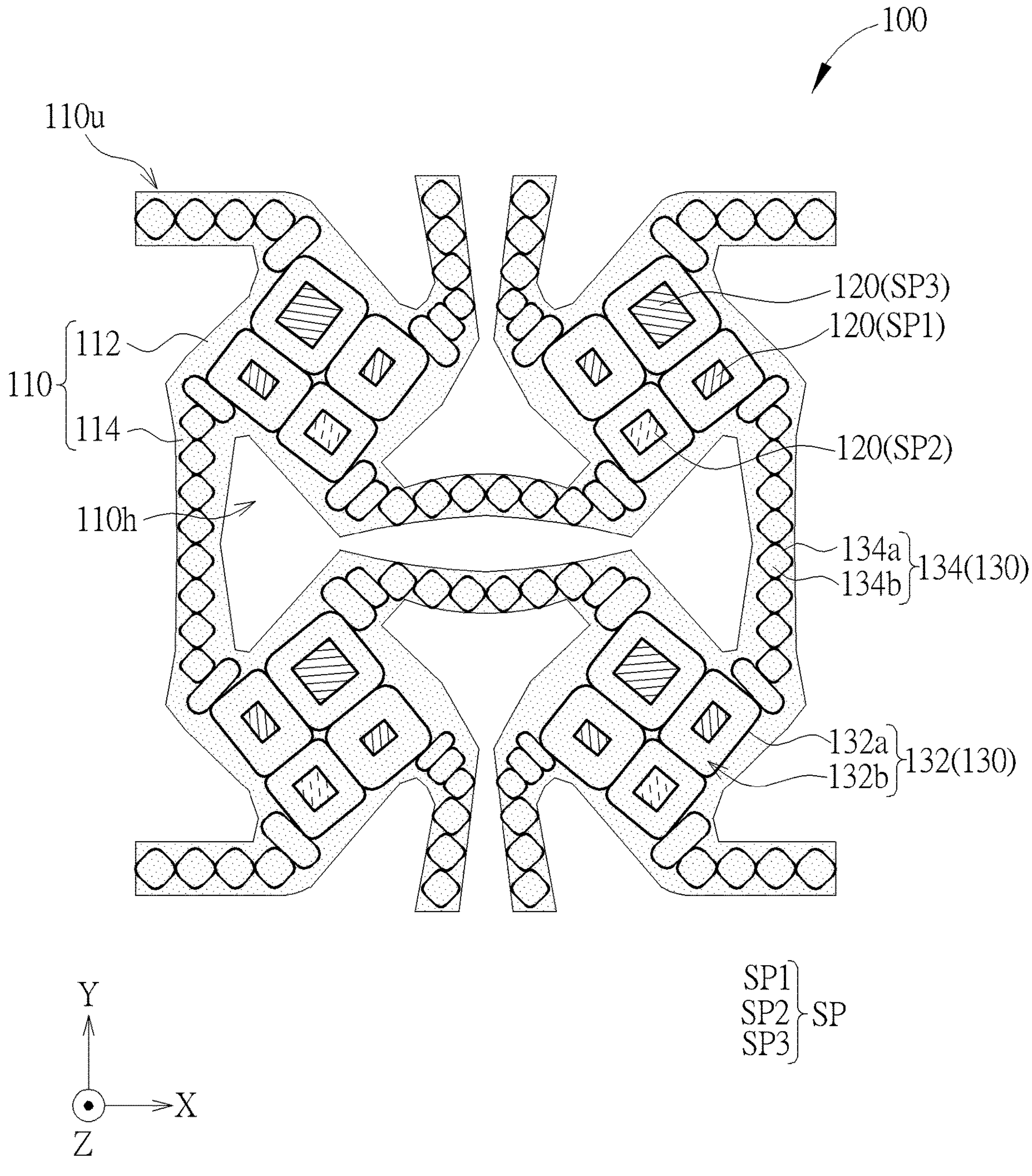
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FIG. 2

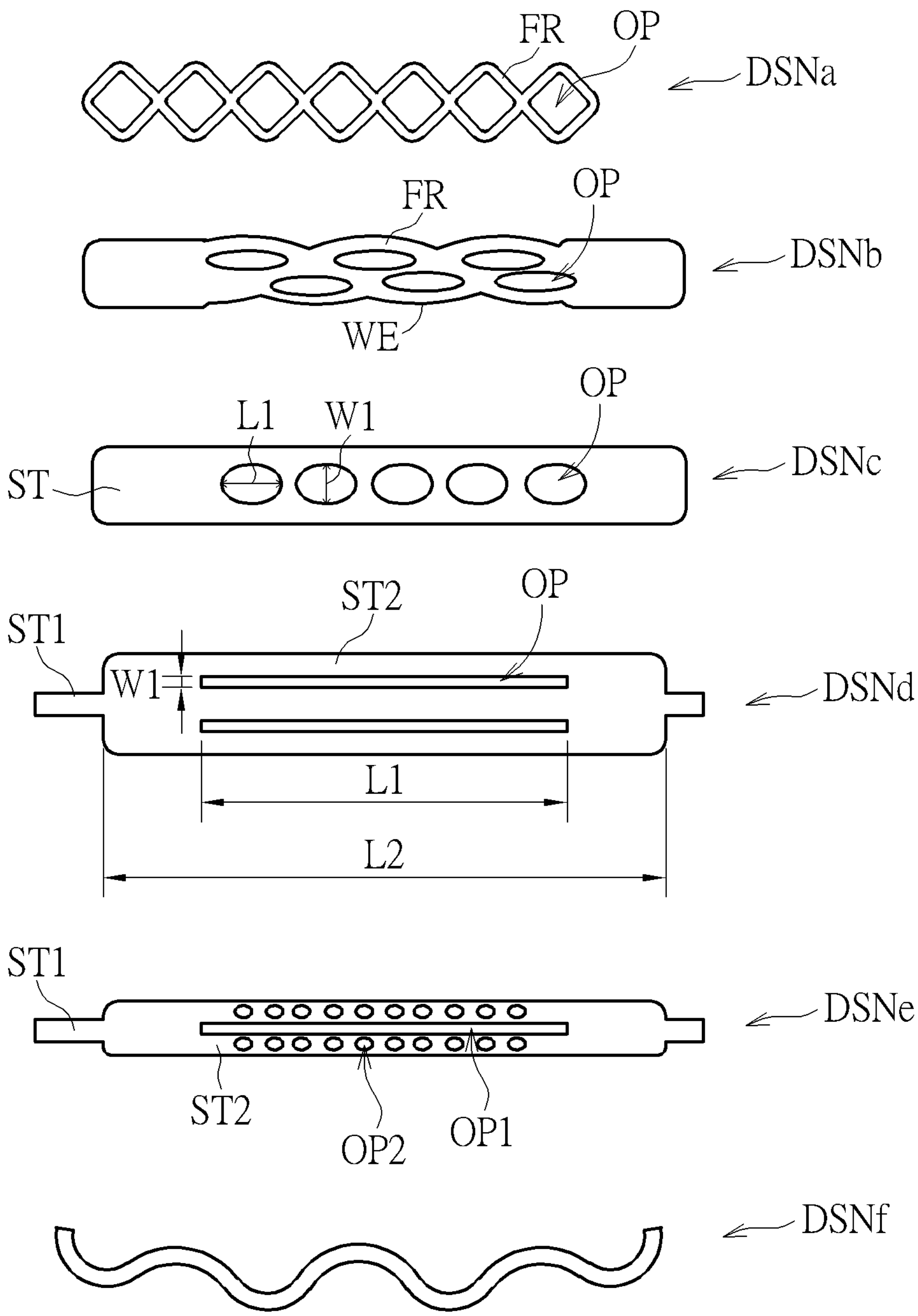
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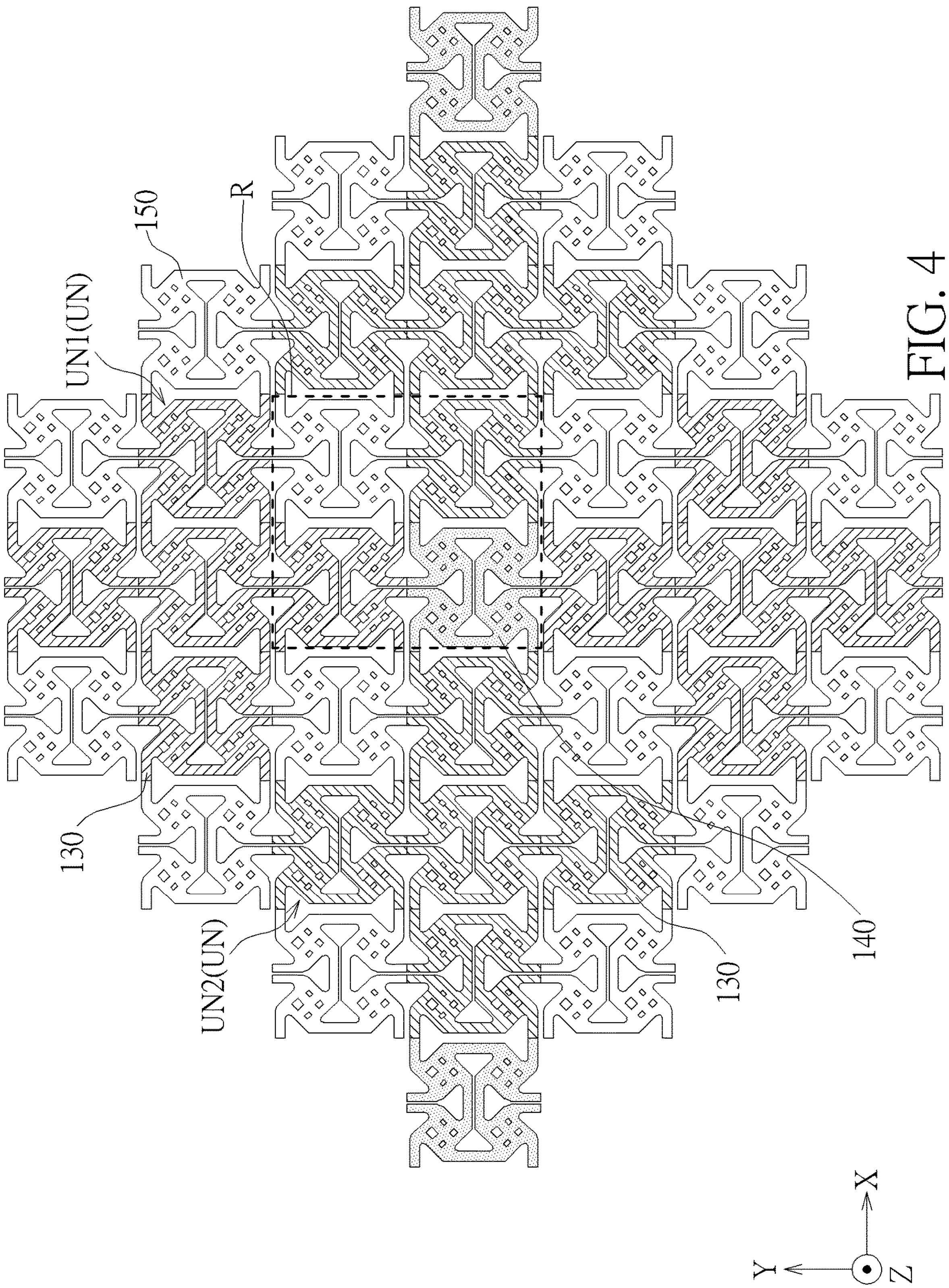
FIG. 3



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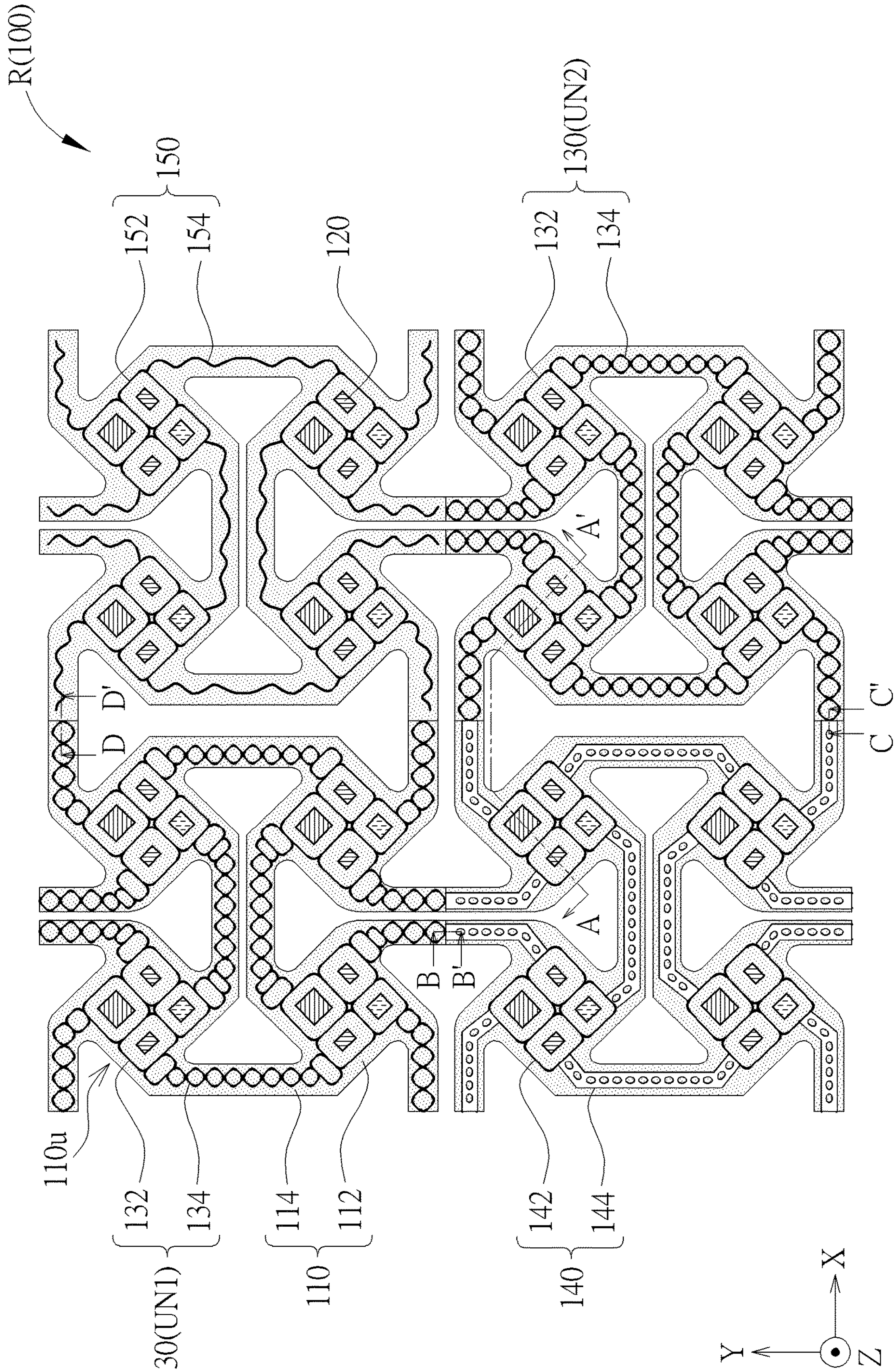
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FIG. 5

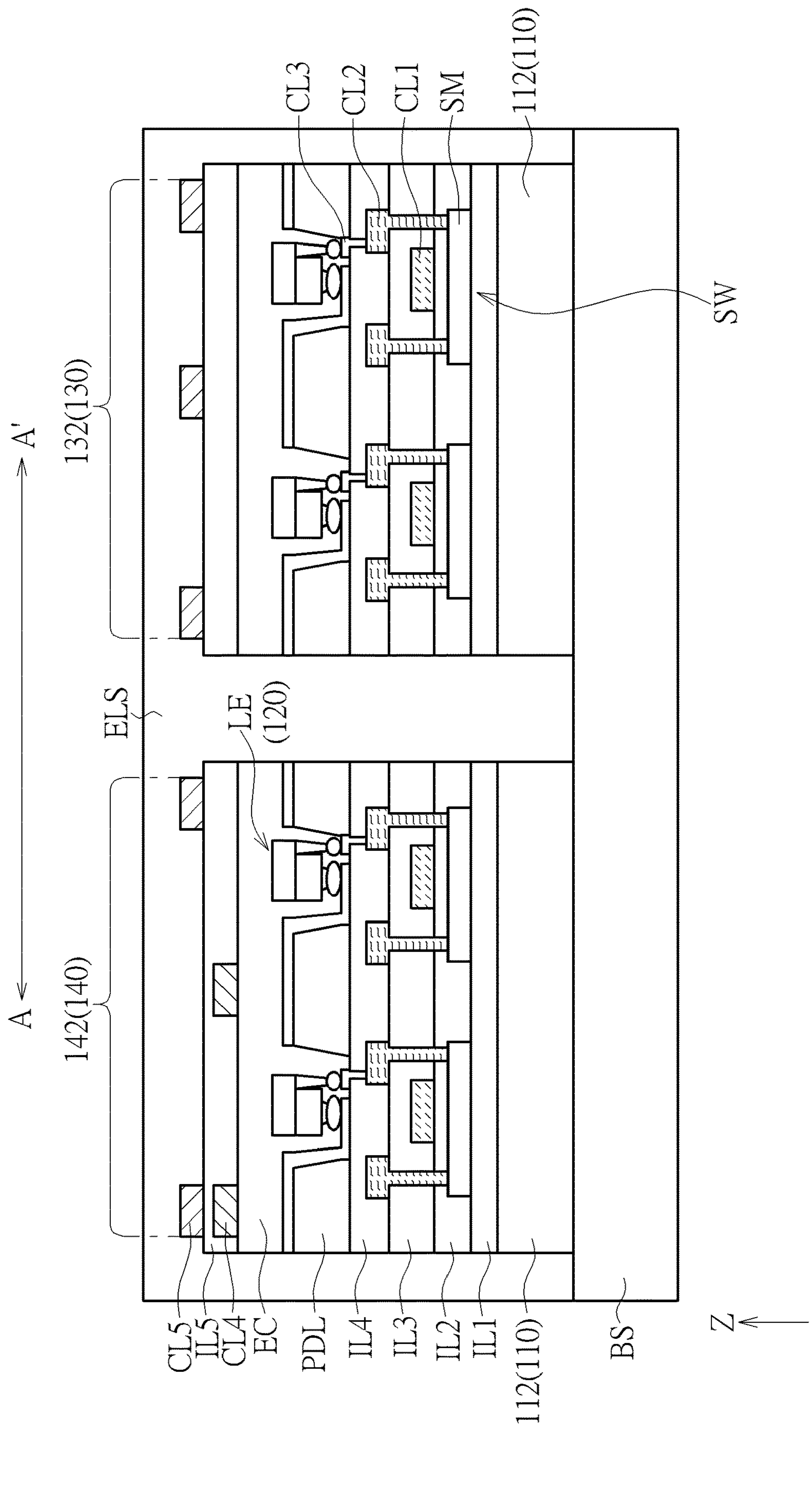

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FIG. 6

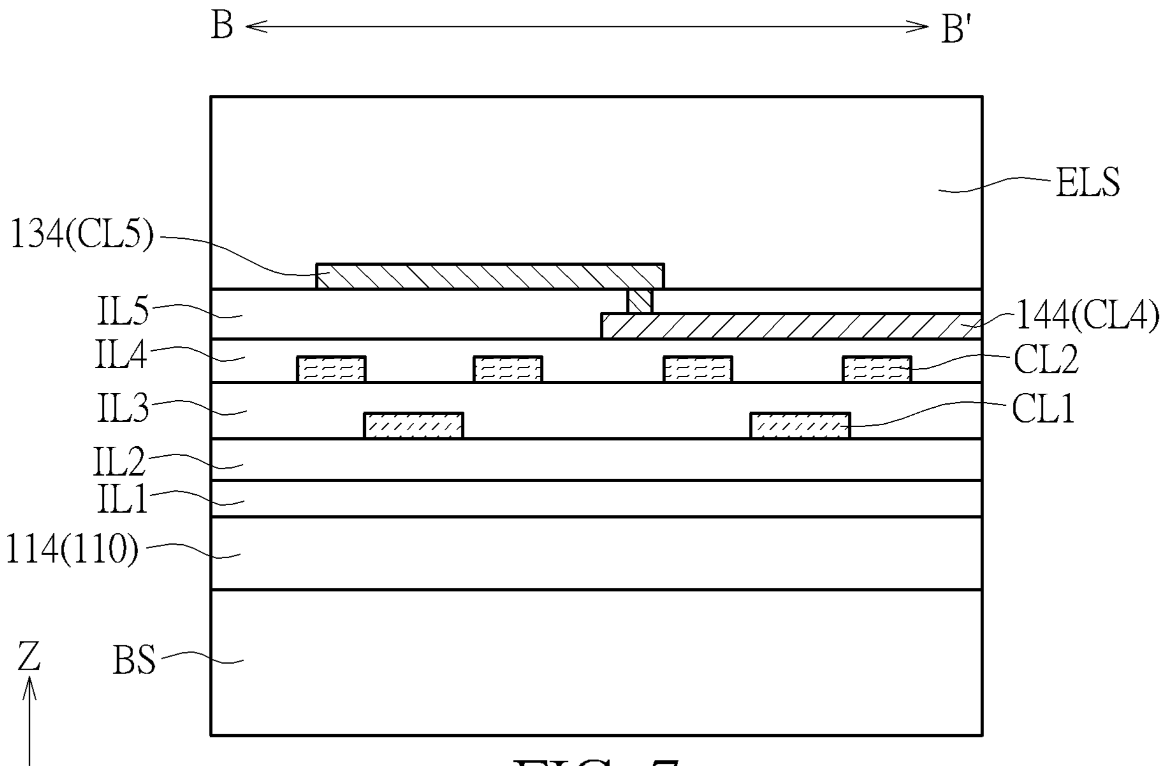
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FIG. 7

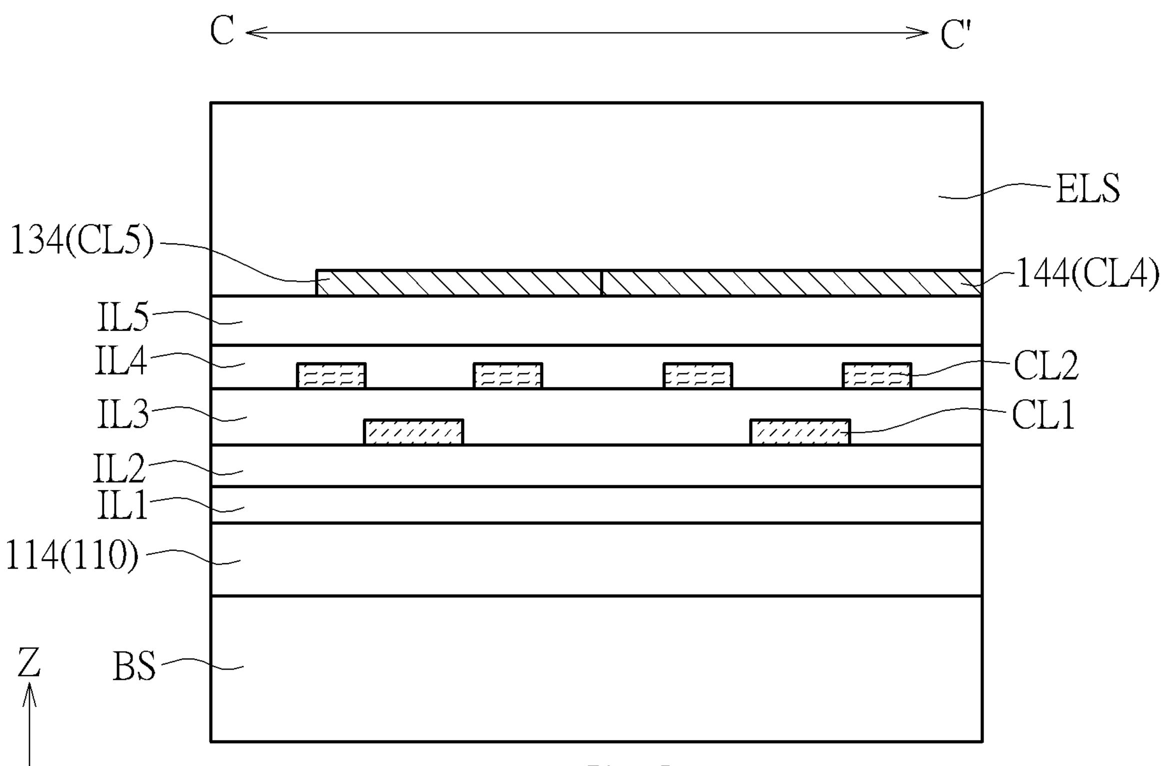
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FIG. 8

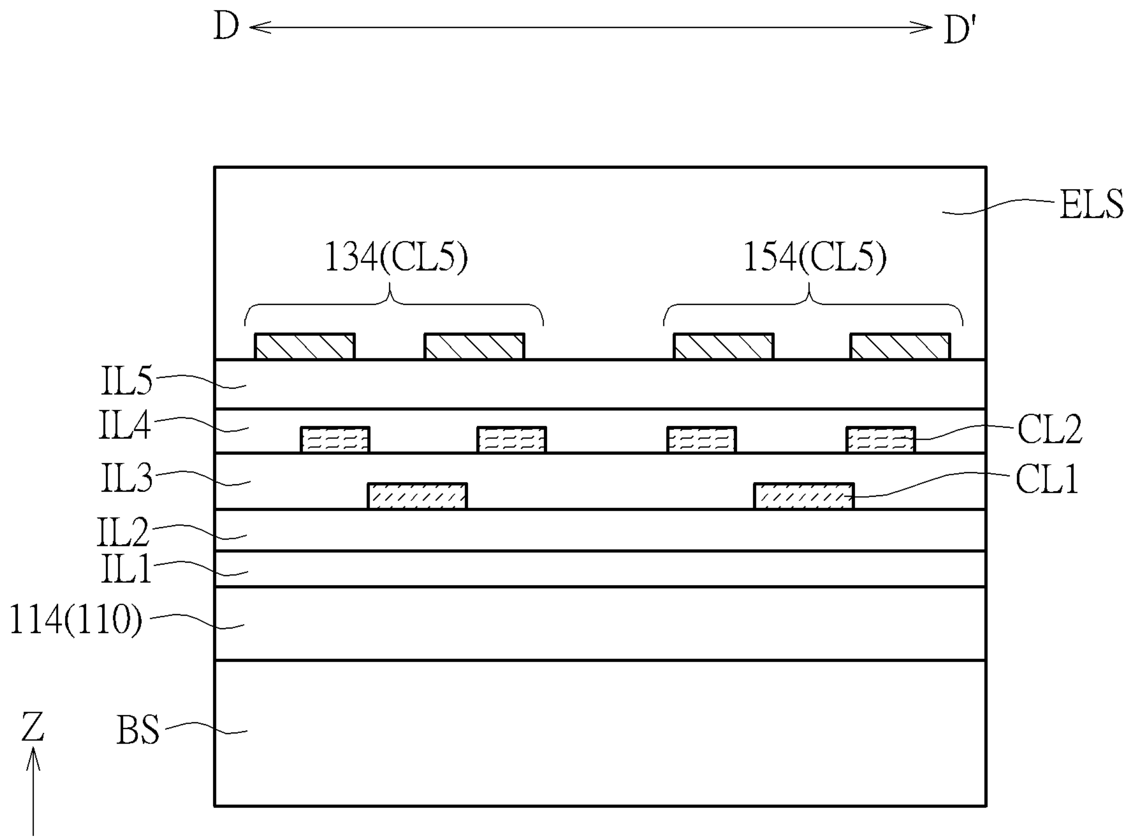

FIG. 9

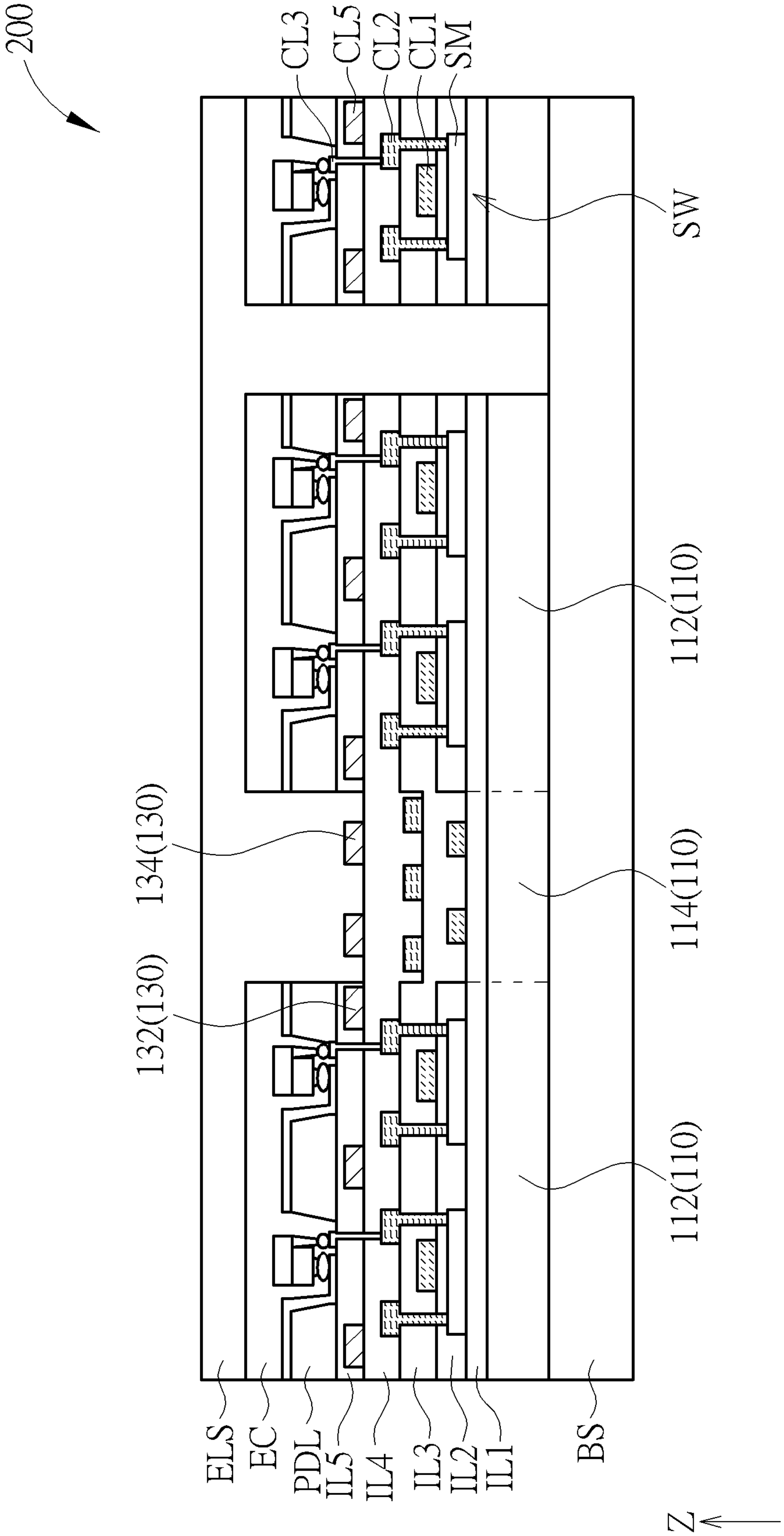
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FIG. 10

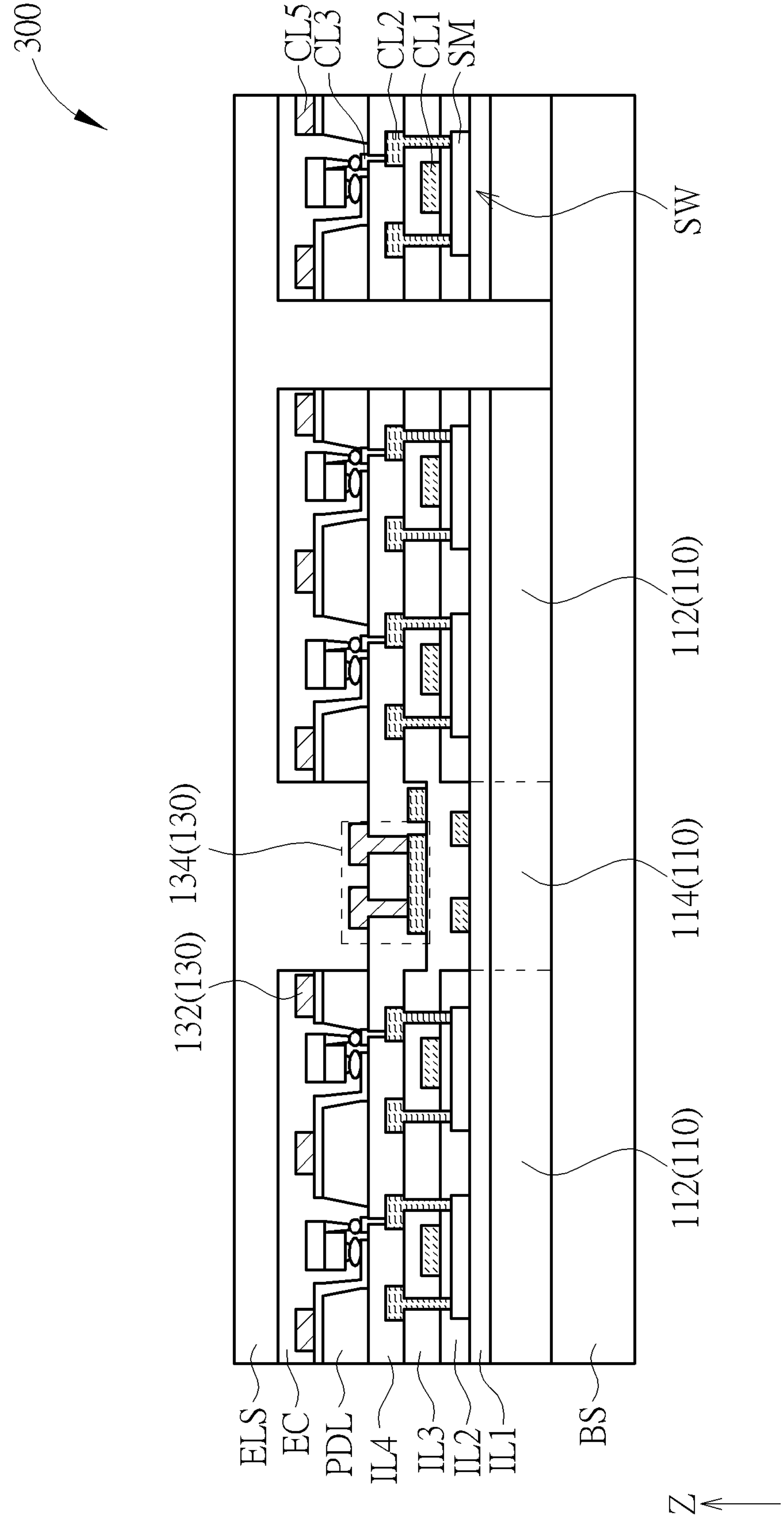
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FIG. 11

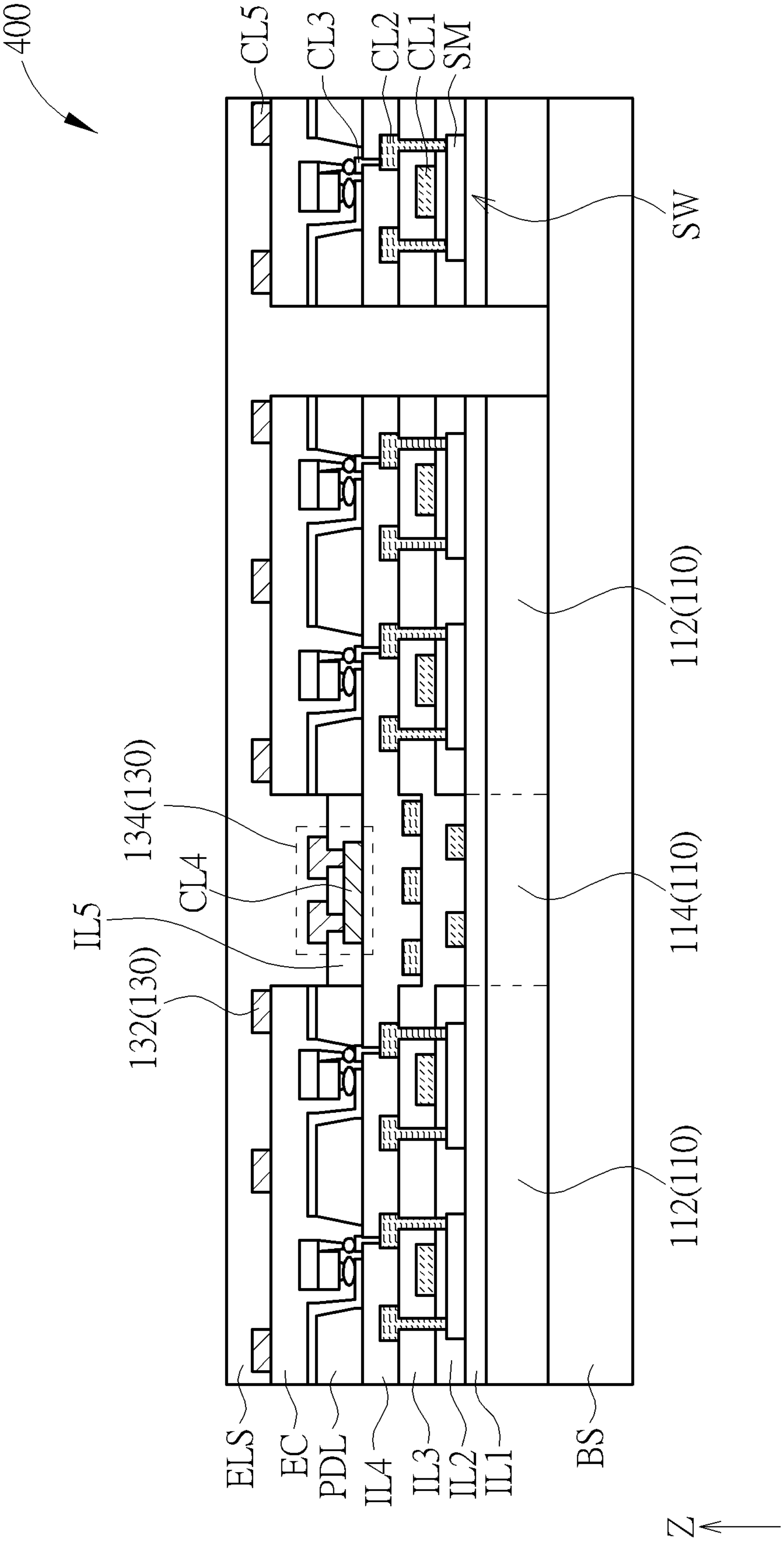
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FIG. 12

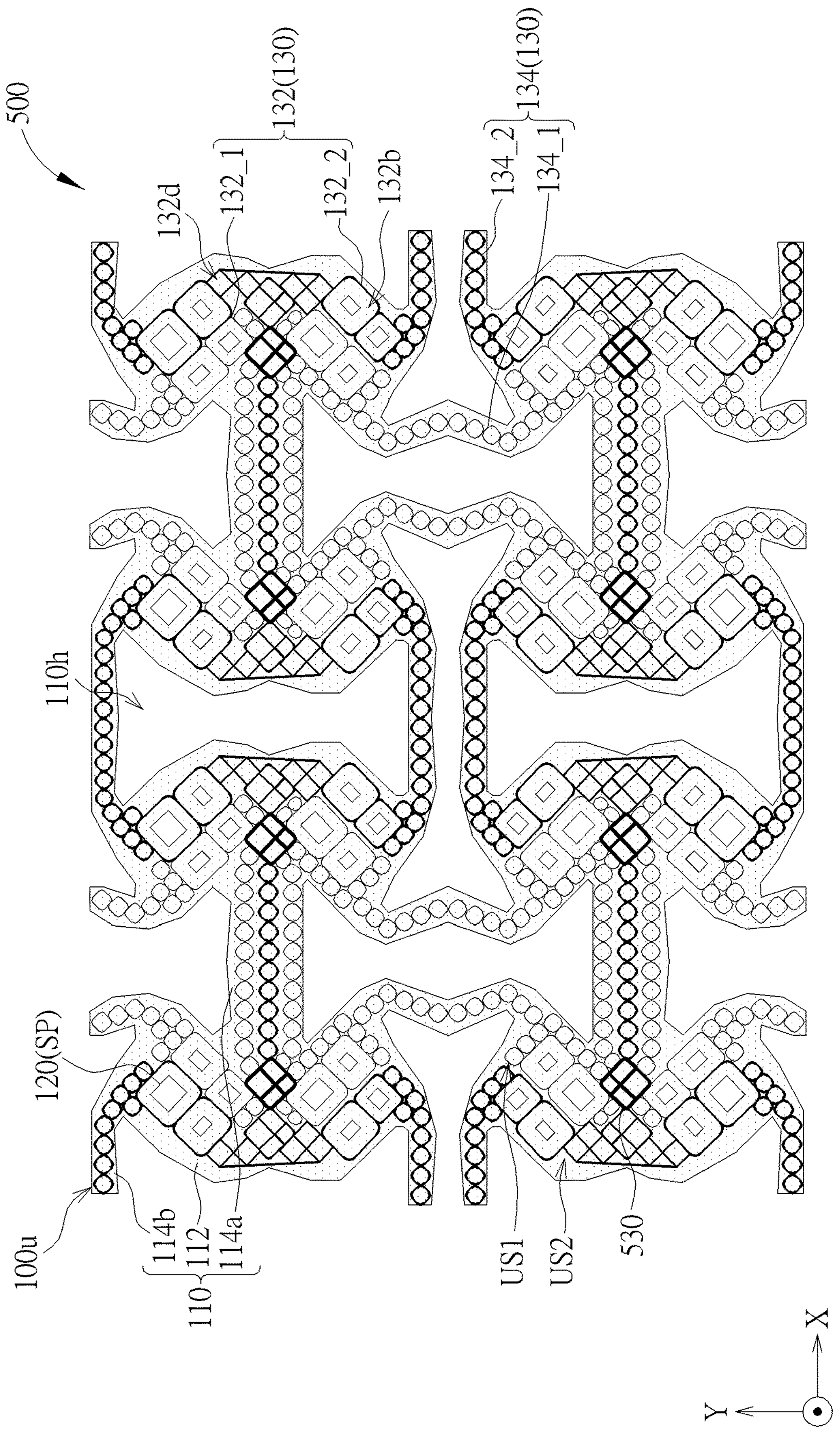
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FIG. 13

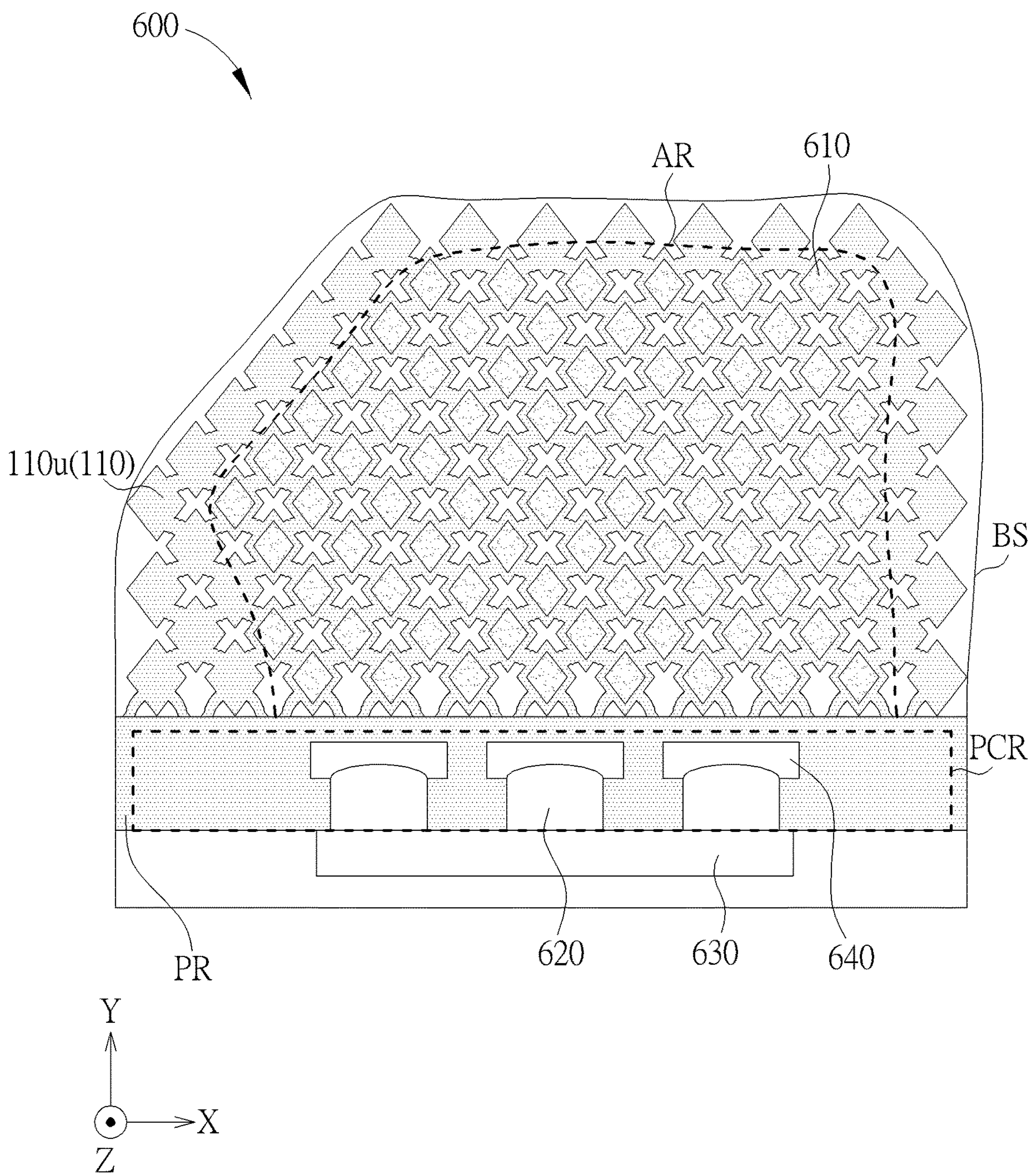

FIG. 14

ELECTRONIC DEVICE

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

The present disclosure relates to an electronic device, and more particularly to an electronic device having a stretchable substrate.

2. Description of the Prior Art

As the evolution and development of electronic devices, the electronic devices have become an indispensable item. In order to make the electronic device be more widely used in various locations and situations (for example, to make the electronic device (e.g., a vehicle electronic device) be disposed on a curved surface or on an irregular object or to make the electronic device (e.g., a wearable electronic device) be worn on the human body), the industry is committed to developing a flexible electronic device and a stretchable electronic device to adapt to various locations and situations. In the designs of the flexible electronic device and the stretchable electronic device, since the bending, curling, stretching and other actions of the electronic device will affect the lifetime, the yield rate and the reliability of the electronic device, it is important to design a flexible electronic devices and a stretchable electronic device which have the long lifetime, the high yield rate, the high reliability and/or the multi-function.

SUMMARY OF THE DISCLOSURE

According to an embodiment, the present disclosure provides an electronic device including a stretchable substrate, a plurality of first electronic units and a plurality of second electronic units. The stretchable substrate has a plurality of island portions and a plurality of bridge portions, and each of the bridge portions connecting at least two of the island portions. The first electronic units are disposed on the island portions. The second electronic units are disposed on the stretchable substrate, each of the second electronic units includes a mesh pattern disposed on at least one of the island portions, and the mesh pattern includes a first opening overlapped with at least one of the first electronic units.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** and FIG. **2** are schematic diagrams showing a top view of a portion of an electronic device according to a first embodiment of the present disclosure.

FIG. **3** is a schematic diagram showing a top view of disconnection-preventing designs according to some embodiments of the present disclosure.

FIG. **4** is a schematic diagram showing a top view of a portion of the electronic device according to the first embodiment of the present disclosure.

FIG. **5** is an enlarge diagram illustrating a region R of FIG. **4**.

FIG. **6** is a schematic diagram showing a cross-sectional view of a structure taken along a cross-sectional line A-A' in FIG. **5**.

FIG. **7** is a schematic diagram showing a cross-sectional view of a structure taken along a cross-sectional line B-B' in FIG. **5**.

FIG. **8** is a schematic diagram showing a cross-sectional view of a structure taken along a cross-sectional line C-C' in FIG. **5**.

FIG. **9** is a schematic diagram showing a cross-sectional view of a structure taken along a cross-sectional line D-D' in FIG. **5**.

FIG. **10** is a schematic diagram showing a cross-sectional view of an electronic device according to a second embodiment of the present disclosure.

FIG. **11** is a schematic diagram showing a cross-sectional view of an electronic device according to a third embodiment of the present disclosure.

FIG. **12** is a schematic diagram showing a cross-sectional view of an electronic device according to a fourth embodiment of the present disclosure.

FIG. **13** is a schematic diagram showing a cross-sectional view of an electronic device according to a fifth embodiment of the present disclosure.

FIG. **14** is a schematic diagram showing a cross-sectional view of an electronic device according to a sixth embodiment of the present disclosure.

DETAILED DESCRIPTION

The present disclosure may be understood by reference to the following detailed description, taken in conjunction with the drawings as described below. It is noted that, for purposes of illustrative clarity and being easily understood by the readers, various drawings of this disclosure show a portion of a display device in this disclosure, and certain elements in various drawings may not be drawn to scale. In addition, the number and dimension of each device shown in drawings are only illustrative and are not intended to limit the scope of the present disclosure.

Certain terms are used throughout the description and following claims to refer to particular components. As one skilled in the art will understand, electronic equipment manufacturers may refer to a component by different names. This document does not intend to distinguish between components that differ in name but not function. In the following description and in the claims, the terms “include”, “comprise” and “have” are used in an open-ended fashion, and thus should be interpreted to mean “include, but not limited to . . . ”. Thus, when the terms “include”, “comprise” and/or “have” are used in the description of the present disclosure, the corresponding features, areas, steps, operations and/or components would be pointed to existence, but not limited to the existence of one or a plurality of the corresponding features, areas, steps, operations and/or components.

The directional terms used throughout the description and following claims, such as: “on”, “up”, “above”, “down”, “below”, “front”, “rear”, “back”, “left”, “right”, etc., are only directions referring to the drawings. Therefore, the directional terms are used for explaining and not used for limiting the present disclosure. Regarding the drawings, the drawings show the general characteristics of methods, structures, and/or materials used in specific embodiments. However, the drawings should not be construed as defining or limiting the scope or properties encompassed by these embodiments. For example, for clarity, the relative size, thickness, and position of each layer, each area, and/or each structure may be reduced or enlarged.

When the corresponding component such as layer or area is referred to “on another component”, it may be directly on this another component, or other component (s) may exist between them. On the other hand, when the component is referred to "directly on another component (or the variant thereof)", any component does not exist between them. Furthermore, when the corresponding component is referred to "on another component", the corresponding component and the another component have a disposition relationship along a top-view/vertical direction, the corresponding component may be below or above the another component, and the disposition relationship along the top-view/vertical direction are determined by an orientation of the device.

It will be understood that when a component or layer is referred to as being "connected to" another component or layer, it can be directly connected to this another component or layer, or intervening components or layers may be presented. In contrast, when a component is referred to as being "directly connected to" another component or layer, there are no intervening components or layers presented. In addition, when the component is referred to "be coupled to/with another component (or the variant thereof)", it may be directly connected to this another component, or may be indirectly connected (such as electrically connected) to this another component through other component(s).

In the description and following claims, the term. "horizontal direction" generally means a direction parallel to a horizontal surface, the term "horizontal surface" generally means a surface parallel to a direction X and direction Y in the drawings, the term "vertical direction" generally means a direction parallel to a direction Z and perpendicular to the horizontal direction in the drawings, and the direction X, the direction Y and the direction Z are perpendicular to each other. In the description and following claims, the term "top view" generally means a viewing result viewing along the vertical direction. In the description and following claims, the term "cross-sectional view" generally means a viewing result viewing a structure cutting along the vertical direction and viewing along the horizontal direction.

In the description and following claims, it should be noted that the term "overlap" means that two elements overlap along the direction Z, and the term "overlap" can be "partially overlap" or "completely overlap" in unspecified circumstances. In the description and following claims, the term "parallel" means that an angle between two elements is less than or equal to the specific degree(s), such as 5 degrees, 3 degrees or 1 degree.

In the description and following claims, in a view (e.g., a top view), the term "surround" means that an element would completely surround or partially surround another element in unspecified circumstances. When "an element B completely surrounds an element C", whole outer edge(s) of the element C would be corresponding to the element B in its/their normal direction. When "an element B partially surrounds an element C", a cyclic structure of the element B configured to surround the element C has at least one gap, such that a part of the outer edge of the element C would not be corresponding to the element B in its normal direction.

The terms "about", "substantially", "equal", or "same" generally mean within ±20% of a given value or range, or mean within ±10%, ±5%, ±3%, ±2%, ±1%, or ±0.5% of a given value or range.

Although terms such as first, second, third, etc., may be used to describe diverse constituent elements, such constituent elements are not limited by the terms. These terms are used only to discriminate a constituent element from other constituent elements in the specification, and these terms have no relation to the manufacturing order of these constituent components. The claims may not use the same terms, but instead may use the terms first, second, third, etc. with respect to the order in which an element is claimed. Accordingly, in the following description, a first constituent element may be a second constituent element in a claim.

It should be noted that the technical features in different embodiments described in the following can be replaced, recombined, or mixed with one another to constitute another embodiment without departing from the spirit of the present disclosure.

According to an embodiment of the present disclosure, a width of the component, a thickness of the component, a height of the component, an area of the component or a distance between two components may be measured by an optical microscopy (OM), a scanning electron microscope (SEM), an Alpha step (α-step), ellipsometer or other suitable method, but not limited thereto. In detail, according to some embodiments, a cross-sectional image containing the component(s) desiring to be measured may be obtained by using the SEM, such that the width of the component, the thickness of the component, the height of the component, the area of the component or the distance between two components may be measured, and a volume of the component may be obtained by a suitable method (e.g., integral). In addition, any two values or directions used for comparison may have certain errors.

In the present disclosure, the electronic device may include a display device, a backlight device, an antenna device, a sensing device or a tiled device, but not limited thereto. The electronic device may be a foldable electronic device, a flexible electronic device or a stretchable electronic device. The display device may be a non-self-luminous type display device or a self-luminous type display device, the antenna device may be a liquid-crystal-type antenna device or a non-liquid-crystal-type antenna device, and the sensing device may be a device for sensing capacitance, light, thermal or ultrasonic, but not limited thereto. Electronic components may include passive component (s) and active component (s), such as capacitors, resistors, inductors, diodes and/or transistors. The diode may include a light emitting diode (LED) or a photodiode. The light emitting diode may include an organic light-emitting diode (OLED), a mini LED, a micro-LED, a quantum-dot LED (QLED, QDLED), but not limited thereto. The electronic device may include liquid crystal material, fluorescence material, phosphor material, quantum dot (QD) material or other suitable material based on requirement(s), but not limited thereto. Furthermore, the display device may be a color display device or a monochrome display device. The electronic device may have a peripheral system (such as a driving system, a control system, a light system, etc.) for supporting the display device, the backlight device, the antenna device, the sensing device or the tiled device. The tiled device may be a tiled display device or a tiled antenna device, but not limited thereto. A shape of the electronic device may be a polygon (e.g., a rectangle), a shape having a curved edge (e.g., a circle, an oval) or other suitable shape, but not limited thereto. Note that the electronic device may be any combination of the above, but not limited thereto. Hereinafter, the stretchable color display device will be used as the stretchable electronic device or the tiled device to illustrate the present disclosure, but the present disclosure is not limited thereto.

In the present disclosure, since the electronic device is stretchable, the electronic device may be used in a variety of situations. In some embodiments, the electronic device may be disposed on a curved surface or on an irregular object. For instance, the electronic device may be a vehicle electronic device disposed on such as windshield, window, rear mirror, instrument panel or steering wheel, but not limited thereto.

In some embodiments, the electronic device may be used in situations that require repeated stretching, wherein the electronic device may be such as a device with a push button and/or a wearable electronic device, but not limited thereto.

In the present disclosure, the electronic device may include an active region and a peripheral region, wherein the active region may optionally include a display region, a sensing region, a light-emitting region, a transceiver region and/or a working region based on the use of the electronic device, the peripheral region may be disposed on at least one outer side of the active region, and electronic components configured to assist the active region may be disposed in the peripheral region, but not limited thereto. For instance, the active region of the stretchable color display device of the present disclosure may include the display region and the sensing region, but not limited thereto.

Referring to FIG. **1** to FIG. **3**, FIG. **1** and FIG. **2** are schematic diagrams showing a top view of a portion of an electronic device according to a first embodiment of the present disclosure, and FIG. **3** is a schematic diagram showing a top view of disconnection-preventing designs according to some embodiments of the present disclosure, wherein the structure shown in FIG. **1** is not stressed such that this structure is not stretched and deformed, and the structure shown in FIG. **2** is stressed such that this structure is stretched and/or deformed. As shown in FIG. **1**, the electronic device **100** includes a stretchable substrate **110**, wherein the stretchable substrate **110** may include any suitable rigid material and/or flexible material. For example, the stretchable substrate **110** may include glass, quartz, ceramic, sapphire, polyimide (PI), polyethylene terephthalate (PET), other suitable materials or a combination thereof. In the present disclosure, the direction X and the direction Y are perpendicular to a normal direction of the stretchable substrate **110**, and the direction Z is parallel to the normal direction of the stretchable substrate **110**.

In the present disclosure, the stretchable substrate **110** may have a patterned structure, so as to make the stretchable substrate **110** stretchable. Namely, even if the material of the stretchable substrate **110** is not stretchable or flexible (e.g., the rigid material), the stretchable substrate **110** may be stretchable due to the patterned structure. In the manufacture of the stretchable substrate **110**, for example, a patterning process may be performed on a substrate to forma plurality of holes **110***h*, so as to manufacture the patterned structure, thereby forming the stretchable substrate **110**.

In detail, the patterned structure of the stretchable substrate **110** may include a plurality of island portions **112** and a plurality of bridge portions **114**, wherein each bridge portion **114** may be connected at least two of the island portions **112**. For instance, FIG. **1** shows four island portions **112**, each island portion **112** may be connected to four bridge portions **114**, and the bridge portion **114** may be connected between two island portions **112**, but not limited thereto. In the present disclosure, the shape of the island portion **112** and the shape of the bridge portion **114** may be designed based on requirement(s), wherein the shape may be such as a polygon (e.g., a rectangle), a shape having a curved edge (e.g., a circle, an oval) or other suitable shape. For example, in FIG. **1**, the island portion **112** may be a quadrangle, and the bridge portion **114** may be a strip structure, but not limited thereto. In the present disclosure, the arrangement of the island portions **112** and the arrangement of the bridge portions **114** may be designed based on requirement(s). For example, in FIG. **1**, the island portions **112** may be arranged in rows along the direction X and in columns along the direction Y, some bridge portions **114** may extend along the direction X, and some bridge portions **114** may extend along the direction Y, but not limited thereto. Note that, in some embodiments, an edge of the connection between the bridge portion **114** and the island portion **112** may be a curved edge (e.g., a circular arc) for example, so as to improve the reliability of the stretchable substrate **110** when the stretchable substrate **110** is operated (e.g., stretched and/or deformed), but not limited thereto.

In some embodiments, the stretchable substrate **110** may have a plurality of substrate units **110***u*, each substrate unit **110***u* may include at least one island portion **112** and at least one bridge portion **114**, and the number of the island portion(s) **112** included in the substrate unit **110***u* and the number of the bridge portion(s) **114** included in the substrate unit **110***u* may be designed based on requirement(s). For instance, in FIG. **1**, each substrate unit **110***u* may include four island portions **112** arranged in a 2×2 array and twelve bridge portions **114** directly connected to these island portions **112** (four bridge portions **114** are connected between these island portions **112**, and each of one ends of other eight bridge portions **114** is connected to the island portion **112** of another substrate unit **110***u*), but not limited thereto. That is to say, FIG. **1** shows one substrate unit **110***u*, but not limited thereto. In some embodiments (as shown in FIG. **1**), the substrate unit **110***u* may have a symmetrical top-view pattern, but not limited thereto. In some embodiments (as shown in FIG. **1**), four island portions **112** arranged in the 2×2 array and four bridge portions **114** directly connected between these island portions **112** may surround the H-shaped hole **110***h* of the stretchable substrate **110**, and the top view pattern of two ends of the H-shaped hole **110***h* may be similar to a triangle, but not limited thereto.

As shown in FIG. **2**, since the patterned structure including the island portions **112** and the bridge portions **114** exists in the stretchable substrate **110**, the stretchable substrate **110** may be stretched and/or deformed by applying force. For example, in FIG. **2**, in the case of applying force to the stretchable substrate **110**, the island portion **112** may be rotated (e.g., in top view), and the bridge portion **114** may be deformed, so as to make the stretchable substrate **110** be stretched, but not limited thereto.

In some embodiments, pixels configured to display an image may be disposed on the island portions **112**, wherein the pixel may include at least one sub-pixel SP. In some embodiments, if the electronic device **100** is a color display device, one pixel may include a plurality of sub-pixels SP for instance, such as a green sub-pixel, a red sub-pixel and a blue sub-pixel, but not limited thereto. The number and color of the sub-pixel(s) SP included in the pixel may be adjusted based on requirement(s). In some embodiments, if the electronic device **100** is a monochrome display device, one pixel may include one sub-pixel SP for instance, but not limited thereto. For instance, in the electronic device **100** (the stretchable color display device) shown in FIG. **1**, one island portion **112** may have one pixel, and each pixel may include two green sub-pixels SP**1**, one red sub-pixel SP**2** and one blue sub-pixel SP**3**, but not limited thereto.

The electronic device **100** includes a plurality of first electronic units **120** disposed on the island portions **112** of the stretchable substrate **110**, wherein the first electronic unit **120** may include may include an active component, such as a thin film transistor, a light emitting diode, a photodiode or other suitable active component. In some embodiments, the first electronic unit **120** may be the active component disposed in the sub-pixel SP (e.g., the active component may be a light emitting component), and one sub-pixel SP may include one first electronic unit **120** or a plurality of first electronic units **120**, but not limited thereto.

In the present disclosure, the light emitting component (e.g., the light emitting diode) disposed in the sub-pixel SP may provide the light having the corresponding light-intensity based on the received voltage and/or the received current, and the received voltage and/or the received current of the light emitting component may be related to such as a gray level signal (provided by an integrated chip or provided from outer device), thereby displaying the image. Namely, the intensity of the light generated by the light emitting component is related to the gray level of a region of the display image corresponding to this light emitting component.

The color of the light emitted from the light emitting component may be designed based on requirement(s). For example, the color or the light emitted from the light emitting component may be based on the sub-pixel SP where this light emitting component is disposed, wherein the emitting light may be red light, green light or blue light, but not limited thereto. In some embodiments, all of the light emitting component may emit the same color light, and the electronic device **100** may further include a light converting layer (not shown in figures) disposed on the light emitting component, so as to convert (or filter) the light emitted from the light emitting component into another light with different color. The light converting layer may include color filter, quantum dots (QD) material, fluorescence material, phosphorescence material, other suitable material(s) or a combination thereof. For instance, the light emitting component may emit white light, the light converting layer may convert the white light into another color light which the sub-pixel SP needs, such as red light, green light or blue light, but not limited thereto. For instance, the light emitting component may emit blue light, the light converting layer may convert the blue light into another color light which the sub-pixel SP needs, such as red light or green light, or may not convert to maintain the blue light, but not limited thereto.

The photodiode may generate an electrical signal (e.g., voltage or current) according to the received light, and the generated electrical signal may be used to perform related applications, such as power generation, light sensing, etc.

The transistor may be a top gate thin film transistor, a bottom gate thin film transistor, a dual gate thin film transistor or other suitable transistor, but not limited thereto. In some embodiments, the transistor may be electrically connected to the light emitting component (e.g., the light emitting diode) in the sub-pixel SP, so as to control the received voltage and/or the received current of the light emitting component, but not limited thereto.

The electronic device **100** may include a plurality of second electronic units **130** disposed on the stretchable substrate **110**, wherein the second electronic unit **130** may include a passive component, such as a resistor, a capacitor, an inductor, a trace, an electrode (e.g., a sense electrode) or other suitable passive component. In some embodiments, the second electronic unit **130** may include may include the touch electrode and/or the trace (e.g., a touch sensing trace), such that the second electronic unit **130** may be configured to sense a touch object (e.g., a finger, a stylus pen, etc.) touching the electronic device **100**, so as to be any suitable touch sensor (e.g., a capacitive touch sensor or a resistive touch sensor), but not limited thereto. In some embodiments, the second electronic unit **130** may include a signal transceiver component (e.g., an antenna component), such that the second electronic unit **130** may be configured to receive or emit the electromagnetic wave, so as to make the electronic device **100** have an antenna function, but not limited thereto.

In the present disclosure, the second electronic unit **130** may be a single-layer structure or a multi-layer structure. In some embodiments, the second electronic unit **130** may be a single-layer structure including one conductive layer, but not limited thereto. In some embodiments, the second electronic unit **130** may be a multi-layer structure including a plurality of conductive layers and an insulating layer disposed between two conductive layers, but not limited thereto. In some embodiments, a portion of the second electronic unit **130** may be a single-layer structure, another portion of the second electronic unit **130** may be a multi-layer structure, but not limited thereto. In the present disclosure, the material of the conductive layer may include such as metal, transparent conductive material (such as indium tin oxide (ITO), indium zinc oxide (IZO), etc.), other suitable conductive materials or a combination thereof, and the material of the insulating layer may include such as silicon oxide (SiO_x), silicon nitride (SiN_y), silicon oxynitride (SiO_xN_y), organic insulating material (e.g., photosensitive resin), other suitable insulating materials or a combination thereof, but not limited thereto. For instance, the material of the conductive layer included in the second electronic unit **130** may be metal, but not limited thereto. In addition, the first electronic unit **120** and the second electronic unit **130** may be disposed on the same side of the stretchable substrate **110** or different sides of the stretchable substrate **110** based on requirement(s).

In the present disclosure, the disposing range of each second electronic unit **130** may be designed based on requirement(s). For instance, in FIG. **1**, there is a one-to-one correspondence between the second electronic unit **130** and substrate unit **110***u* of the stretchable substrate **110** (i.e., one second electronic unit **130** is disposed on one substrate unit **110***u*), but not limited thereto.

In some embodiments, the pattern of the second electronic unit **130** may have a disconnection-preventing design based on requirement(s), so as to reduce the disconnection possibility of the second electronic unit **130** when the electronic device **100** is stretched and/or deformed, thereby increasing the reliability of the electronic device **100** when the electronic device **100** is stretched and/or deformed. In actual design, the pattern of the second electronic unit **130** may be designed based on the stretching amount, the deformation degree and/or the bending curvature of the electronic device **100**.

FIG. **3** shows some embodiments of the disconnection-preventing design, but the disconnection-preventing design of the present disclosure is not limited thereto. The disconnection-preventing design may be a combination of the designs shown in FIG. **3** or other suitable design. In some embodiments (as shown in FIG. **3**), the pattern of the disconnection-preventing design may include an opening OP, and the opening OP may be in a shape of a polygon (e.g., a rectangle, a rhombus, a bar), a shape having a curved edge (e.g., a circle, an oval) or other suitable shape. In some embodiments, the shape of the opening OP may be a polygon with chamfers, wherein the existence of the chamfer may enhance the reliability of the electronic device **100** when the electronic device **100** is stretched and/or deformed. In the present disclosure, an area of the opening OP may be designed based on requirement(s).

For example, in the disconnection-preventing design DSNa shown in FIG. **3**, the disconnection-preventing design DSNa may include a frame FR surrounding the opening OP (e.g., the opening OP may have, but not limited to, a rectangle pattern), and the disconnection-preventing design DSNa may be formed by connecting the frames FR in series (e.g., the opening OP may be arranged in a row), but not limited thereto. For example, in the disconnection-preventing design DSNb shown in FIG. **3**, the disconnection-preventing design DSNb may include a frame FR surrounding the opening OP (e.g., the opening OP may have, but not limited to, an oval pattern), the frames FR may be connected to each other to make the disconnection-preventing design DSNb have a wavy outer edge WE, the opening OP may be arranged in rows (the disconnection-preventing design DSNb of FIG. **3** has two rows of the opening OP), and the openings OP in different rows may be staggered, but not limited thereto. For example, in the disconnection-preventing design DSNc shown in FIG. **3**, the disconnection-preventing design DSNc may be a strip structure ST including at least one opening OP (e.g., the opening OP may have, but not limited to, a circle pattern or an oval pattern), wherein the opening OP may be arranged in at least one row. In some embodiments (e.g., the disconnection-preventing design DSNa, the disconnection-preventing design DSNb and the disconnection-preventing design DSNc), the area of the opening OP may be greater than or equal to 1 μm^2 and less than or equal to 50 μm^2 (i.e., 1 $\mu m^2 \leq$ the area of the opening OP$\leq$50 μm^2), so as to increase the reliability of the electronic device **100** when the electronic device **100** is stretched and/or deformed, but not limited thereto. In some embodiments (e.g., the disconnection-preventing design DSNa, the disconnection-preventing design DSNb and the disconnection-preventing design DSNc), in one opening OP, a ratio of a length L**1** to a width W**1** may be greater than or equal to 1 and less than or equal to 5 (i.e., 1≤L**1**/W**1**≤5), so as to increase the reliability of the electronic device **100** when the electronic device **100** is stretched and/or deformed, but not limited thereto. Note that, in the present disclosure, the length L**1** is measured along a direction parallel to an extension direction (a length direction) of the strip structure ST, and the width W**1** is measured along a direction perpendicular to the extension direction (the length direction) of the strip structure ST.

In some embodiments, the disconnection-preventing design DSNd and the disconnection-preventing design DSNe shown in FIG. **3** may have a forking structure, so as to make one strip pattern ST**1** be divided into a plurality of branched strip patterns ST**2**. In the disconnection-preventing design DSNd, one strip pattern ST**1** may be divided into three branched strip patterns ST**2**, and the opening OP (e.g., the opening OP may have a rectangle pattern or a bar pattern) may be disposed between two adjacent branched strip patterns ST**2** among in the three branched strip patterns ST**2**, but not limited thereto. In the disconnection-preventing design DSNe, one strip pattern ST**1** may be divided into two branched strip patterns ST**2**, and the opening OP**1** (e.g., the opening OP**1** may have a rectangle pattern or a bar pattern) may be disposed between two branched strip patterns ST**2**. Since the disconnection-preventing design DSNd and the disconnection-preventing design DSNe have the forking structure, even if one of the branched strip patterns ST**2** breaks during the stretching process of the electronic device **100**, the electrical signal may still be transmitted through other branched strip pattern(s) ST**2**, thereby decreasing the possibility of disconnection when the electronic device **100** is stretched and/or deformed. In some embodiments (e.g., the disconnection-preventing design DSNd and the disconnection-preventing design DSNe), a ratio of a length L**2** of the branched strip pattern ST**2** to a length L**1** of the opening OP (or opening OP**1**) may be greater than or equal to 1 and less than or equal to 2 (i.e., 1≤L**2**/L**1**≤2), so as to increase the reliability of the electronic device **100** when the electronic device **100** is stretched and/or deformed, but not limited thereto. In some embodiments (e.g., the disconnection-preventing design DSNd and the disconnection-preventing design DSNe), in one opening OP (or opening OP**1**), a ratio of the length L**1** to a width W**1** may be greater than or equal to 50 and less than or equal to 1000 (i.e., 50≤L**1**/W**1**≤1000), so as to increase the reliability of the electronic device **100** when the electronic device **100** is stretched and/or deformed, but not limited thereto. Note that, in the present disclosure, the length L**1** is measured along a direction parallel to an extension direction (a length direction) of the branched strip structure ST**2**, and the width W**1** is measured along a direction perpendicular to the extension direction (the length direction) of the branched strip structure ST**2**.

Furthermore, the disconnection-preventing design may have multiple types of openings. For instance, the disconnection-preventing design DSNe may have the opening OP**1** having a bar pattern (or a rectangle pattern) and the opening OP**2** having a circle pattern, and the opening OP**2** having the circle pattern may be disposed in the branched strip pattern ST**2**, so as to increase the reliability of the electronic device **100** when the electronic device **100** is stretched and/or deformed, but not limited thereto. In addition, the disconnection-preventing design may optionally have a gap without affecting the transmission of the electrical signal, so as to improve the disconnection-preventing effect.

In some embodiments (as shown in FIG. **3**), the pattern of the disconnection-preventing design DSNf may have a wavy shape, so as to enhance the stretchable property of the disconnection-preventing design DSNf, thereby increasing the reliability of the electronic device **100** when the electronic device **100** is stretched and/or deformed, but not limited thereto.

In the present disclosure, each second electronic unit **130** may include a mesh pattern **132** disposed on the island portion **112**, wherein the mesh pattern **132** may optionally have the disconnection-preventing design (i.e., the mesh pattern **132** may include one of the disconnection-preventing designs shown in FIG. **3**, a combination of the disconnection-preventing designs shown in FIG. **3** or other suitable disconnection-preventing design), so as to decrease the possibility of disconnection of the mesh pattern **132** when the electronic device **100** is stretched and/or deformed, thereby increasing the reliability of the electronic device **100** when the electronic device **100** is stretched and/or deformed. In FIG. **1**, the mesh pattern **132** may include a first frame **132*a*** and a first opening **132*b***, the first frame **132*a*** may surround (completely surround or partially surround) the first opening **132*b***, and the first opening **132*b*** may overlap at least one first electronic unit **120**. Therefore, in FIG. **1**, the first frame **132*a*** may surround at least one first electronic unit **120**, and the first frame **132*a*** may not overlap the first electronic unit **120**. In the present disclosure, the first frame **132*a*** of the second electronic unit **130** may have a closed pattern or a non-closed pattern based on requirement(s), so as to completely surround or partially surround the first opening **132*b***. In the present disclosure, the first frame **132*a*** of the mesh pattern **132** of the second electronic unit **130** may include a conductive material (e.g., the conductive material may be, but not limited to, metal). In the present disclosure, the shape of the first frame **132*a*** and the shape of the first opening **132*b*** may be designed to be any suitable shape based on requirement(s). For example, the shape of the first frame **132*a*** and the shape of the first opening **132*b*** may be a polygon (e.g., a rectangle, a rhombus, a bar), a shape having a curved edge (e.g., a circle, an oval), but not limited thereto. In some embodiments, the shape of the first frame **132***a* and the shape of the first opening **132***b* may be a polygon with chamfers, so as to increase the reliability of the electronic device **100** when the electronic device **100** is stretched and/or deformed.

In the present disclosure, the distribution of the mesh pattern(s) **132** of the second electronic unit **130** on the substrate unit **110***u* may be designed based on requirement (s), such that the mesh pattern(s) **132** may be disposed on one island portion **112** or a plurality of island portions **112** of the substrate unit **110***u*. In some embodiments, the mesh patterns **132** of the second electronic unit **130** may be disposed on at least two island portions **112** of the substrate unit **110***u*. For example, in FIG. **1**, the mesh patterns **132** of the second electronic unit **130** may be disposed on all island portions **112** of the substrate unit **110***u*, but not limited thereto. Moreover, one mesh pattern **132** or a plurality of mesh patterns **132** may be disposed on one island portion **112**. For example, in FIG. **1**, four mesh patterns **132** may be disposed on one island portion **112**, but not limited thereto.

Each second electronic unit **130** may include a connection pattern **134** disposed on the bridge portion **114**, wherein the connection pattern **134** may be configured to be electrically connected to the mesh patterns **132** on two adjacent island portions **112**, and the pattern of the connection pattern **134** may be designed based on requirement(s). In some embodiments, the pattern of the connection pattern **134** may have the disconnection-preventing design (i.e., the connection pattern **134** may include one of the disconnection-preventing designs shown in FIG. **3**, a combination of the disconnection-preventing designs shown in FIG. **3** or other suitable disconnection-preventing design), so as to decrease the possibility of disconnection of the connection pattern **134** when the electronic device **100** is stretched and/or deformed, thereby increasing the reliability of the electronic device **100** when the electronic device **100** is stretched and/or deformed.

In some embodiments (as shown in FIG. **1**), the connection pattern **134** may include a second frame **134***a* and a second opening **134***b*, wherein the shape of the second opening **134***b* may be a polygon (e.g., a rectangle, a rhombus, a bar), a shape having a curved edge (e.g., a circle, an oval) or other suitable shape, and the second frame **134***a* may surround (completely surround or partially surround) the second opening **134***b*. In the present disclosure, the second frame **134***a* may have a closed pattern or a non-closed pattern based on requirement (s), so as to completely surround or partially surround the second opening **134***b*. For instance, the shape of the second frame **134***a* and the shape of the second opening **134***b* may be a polygon (e.g., a rectangle) with chamfers, and the connection pattern **134** may be formed by connecting the second frames **134***a* in series, so as to increase the reliability of the electronic device **100** when the electronic device **100** is stretched and/or deformed, but not limited thereto.

Under the condition that the connection pattern **134** has the second opening **134***b*, the area relationship between the first opening **132***b* of the mesh pattern **132** and the second opening **134***b* of the connection pattern **134** may be designed based on requirement(s). In FIG. **1**, the first opening **132***b* has a first area A**1**, the second opening **134***b* has a second area A**2**, and the first area A**1** may be greater than the second area A**2**. For example, a ratio of the first area A**1** to the second area A**2** may be greater than 50 and less than or equal to 5000 (i.e., $50<A1/A2\leq5000$), so as to increase the reliability of the electronic device **100** when the electronic device **100** is stretched and/or deformed, but not limited thereto. Furthermore, optionally, the areas of the first openings **132***b* of the mesh patterns **132** on the same island portion **112** may be the same or different from each other based on requirement(s). For instance, in one island portion **112** shown in FIG. **1**, the greatest first opening **132***b* and the smallest first opening **132***b* (or referred as a third opening) respectively overlaps different first electronic units **120**, and a ratio of an area of the greatest first opening **132***b* (e.g., the first area A**1**) to an area of the smallest first opening **132***b* (or referred as a third area A**3** of the third opening) may be greater than 1 and less than or equal to 5 (i.e., $1<A1/A3\leq5$), but not limited thereto. For example, the areas of the first openings **132***b* of the mesh patterns **132** on the same the island portion **112** may be the same (i.e., the first area A**1** and the third area A**3** are the same), but not limited thereto.

In some embodiments (not shown in figures), the connection pattern **134** may have the wavy shape, so as to enhance the stretchable property of the connection pattern **134**, thereby increasing the reliability of the electronic device **100** when the electronic device **100** is stretched and/or deformed, but not limited thereto.

In some embodiments, the connection patterns **134** located at different positions in one second electronic unit **130** may have different pattern designs. For instance (not shown in figures), in one substrate unit **110***u*, the connection pattern **134** having the disconnection-preventing design DSNa shown in FIG. **3** may be disposed on one bridge portion **114**, the connection pattern **134** having the disconnection-preventing design DSNf shown in FIG. **3** may be disposed on another bridge portion **114**, but not limited thereto. In addition, the connection pattern **134** may be designed based on the stretching amount, the deformation degree and/or the bending curvature of the electronic device **100**. For instance, if the electronic device **100** has the greater stretching amount and/or the greater deformation degree in the direction X, and the electronic device **100** is not stretched and deformed in the direction Y, the connection pattern **134** on the bridge portion **114** extending along the direction X may have the disconnection-preventing design, and the connection pattern **134** on the bridge portion **114** extending along the direction Y may not have the disconnection-preventing design, but not limited thereto.

In the present disclosure, the distribution of the connection pattern(s) **134** of the second electronic unit **130** on the substrate unit **110***u* may be designed based on requirement(s), such that the connection pattern(s) **134** may be disposed on one bridge portion **114** or a plurality of bridge portions **114** of the substrate unit **110***u*. For instance, as shown in FIG. **1**, the connection patterns **134** may be disposed on each bridge portion **114** of the substrate unit **110***u* corresponding to the second electronic unit **130**, but not limited thereto. For instance (not shown in figures), the connection pattern(s) **134** may be disposed on at least one bridge portion **114** of the substrate unit **110***u* corresponding to the second electronic unit **130**, and the connection pattern **134** may not be disposed on at least another one bridge portion **114** of the substrate unit **110***u* corresponding to the second electronic unit **130**, but not limited thereto.

When the force is applied on the stretchable substrate **110** to make the stretchable substrate **110** stretched and/or deformed, at least a portion of the pattern of the second electronic unit **130** may be deformed. For example, in FIG. **2**, when the island portion **112** is rotated due to the applying force, the connection pattern **134** of the second electronic unit **130** is deformed at the connection connected to the mesh pattern **132**. In some embodiments, the deformed portion of the second electronic unit **130** may have a pattern more adaptable to the deformation. For instance, the deformed portion of the second electronic unit **130** may have a greater opening, or the pattern of the deformed portion of the second electronic unit **130** may be designed based on the direction and/or curvature of the deformation of the part of the stretchable substrate **110** corresponding to the deformed portion.

Referring to FIG. **4** to FIG. **9**, and referring to FIG. **1**, FIG. **4** is a schematic diagram showing a top view of a portion of the electronic device according to the first embodiment of the present disclosure, FIG. **5** is an enlarge diagram illustrating a region R of FIG. **4**, and FIG. **6** to FIG. **9** are schematic diagrams showing cross-sectional views of structures taken along cross-sectional lines A-A', B-B', C-C' and D-D' in FIG. **5** respectively, wherein an object BS shown in FIG. **6** to FIG. **9** is any object where the electronic device **100** can be disposed, the object BS may be, but not limited to, windshield, window, rear mirror, instrument panel, steering wheel, animal or human, and the first electronic unit **120** is the light emitting component LE as an example. As shown in FIG. **4**, the electronic device **100** may include a plurality of uniting units UN, and each uniting unit UN may include a plurality of second electronic units **130** connected to each other. For instance, in FIG. **4**, each uniting unit UN may include five second electronic units **130**, and five second electronic units **130** may be connected to each other to form a "+" shape; namely, each uniting unit UN may be disposed on the multiple substrate units **110***u* of the stretchable substrate **110**, but not limited thereto. In some embodiments (as shown in FIG. **4**), the uniting units UN may include a plurality of first uniting units UN**1** and a plurality of second uniting unit UN**2**, and the first uniting unit UN**1** and the second uniting unit UN**2** are not directly connected to each other. For example, if the second electronic unit **130** is configured to perform the touch sensing function, the uniting unit UN may be a sensing electrode, wherein the first uniting unit UN**1** may be such as a signal sending electrode (e.g., Tx electrode), and the second uniting unit UN**2** may be such as a signal receiving electrode (e.g., Rx electrode), such that the uniting units UN including the second electronic units **130** may be a capacitive touch sensor, but not limited thereto.

In the present disclosure, the arrangements and connections of the first uniting units UN**1** and the second uniting units UN**2** may be designed based on requirement(s). For instance, the first uniting units UN**1** arranged along the direction Y may be electrically connected to each other to form an uniting unit column, the uniting unit columns may be arranged along the direction X, the second uniting units UN**2** arranged along the direction X may be electrically connected to each other to form an uniting unit row, and the uniting unit rows may be arranged along the direction Y, but not limited thereto.

The electronic device **100** may further include a plurality of bridge units **140** disposed on the stretchable substrate **110**. For example, in FIG. **4**, there is a one-to-one correspondence between the bridge unit **140** and the substrate unit **110***u* of the stretchable substrate **110** (i.e., one bridge unit **140** is disposed on one substrate unit **110***u*), but not limited thereto. In the present disclosure, each bridge unit **140** may be electrically connected to two adjacent second electronic units **130**, and these two adjacent second electronic units **130** may belong to different uniting units UN respectively. In detail, as shown in FIG. **4**, the bridge unit **140** may be between two adjacent first uniting units UN**1** in the direction Y, so as to be electrically connected to the second electronic units **130** belonging to different first uniting units UN**1** respectively, thereby being electrically connected to two adjacent first uniting units UN**1**; the bridge unit **140** may be between two adjacent second uniting units UN**2** in the direction X, so as to be electrically connected to the second electronic units **130** belonging to different the second uniting units UN**2** respectively, thereby being electrically connected to two adjacent second uniting units UN**2**. Note that, the bridge unit **140** is electrically connected two adjacent first uniting units UN**1** with a first conductive path and electrically connected two adjacent second uniting units UN**2** with a second conductive path, and the first conductive path is different from the second conductive path, such that the first uniting unit UN**1** does not conduct to the second uniting unit UN**2**.

Similarly, the bridge unit **140** may include a mesh pattern **142** disposed on the island portion **112** and a connection pattern **144** disposed on the bridge portion **114** (as shown in FIG. **5**). Details of the mesh pattern **142** and the connection pattern **144** may be referred to the mesh pattern **132** and the connection pattern **134** of the second electronic unit **130** described above, and these parts will not be redundantly described.

Optionally, the electronic device **100** may further include a plurality of dummy units **150** disposed on the stretchable substrate **110**. For instance, in FIG. **4**, there is a one-to-one correspondence between the dummy unit **150** and the substrate unit **110***u* of the stretchable substrate **110** (i.e., one dummy unit **150** is disposed on one substrate unit **110***u*), but not limited thereto. In the present disclosure, the dummy unit **150** may be disposed between and electrically insulated from two adjacent second electronic units **130**, wherein these two adjacent second electronic units **130** may respectively belong to different uniting units UN. In some embodiments, the dummy unit **150** may be configured to separate the first uniting unit UN**1** from the second uniting unit UN**2**.

Optionally, the dummy unit **150** may include a mesh pattern **152** disposed on the island portion **112** and/or a connection pattern **154** disposed on the bridge portion **114** (as shown in FIG. **5**). Details of the mesh pattern **152** and the connection pattern **154** may be referred to the mesh pattern **132** and the connection pattern **134** of the second electronic unit **130** described above, and these parts will not be redundantly described.

FIG. **5** shows four substrate units **110***u* and the structures disposed on them in the region R, wherein the second electronic unit **130** of the first uniting unit UN**1**, the second electronic unit **130** of the second uniting unit UN**2**, the bridge unit **140** and the dummy unit **150** are respectively disposed on these four substrate units **110***u*. In the present disclosure, the mesh pattern **132**, the mesh pattern **142** and the mesh pattern **152** disposed on different substrate units **110***u* may have the same design or different designs based on requirement(s), and the connection pattern **134**, the connection pattern **144** and the connection pattern **154** disposed on different substrate units **110***u* may have the same design or different designs based on requirement(s). In some embodiments (as shown in FIG. **5**), the mesh pattern **132**, the mesh pattern **142** and the mesh pattern **152** disposed on different substrate units **110***u* (disposed on the island portions **112**) may have the same design, and the connection pattern **134**, the connection pattern **144** and the connection pattern **154** disposed on different substrate units **110***u* (disposed on the bridge portions **114**) may have different designs. For instance, in FIG. **5**, the connection pattern **134** of the second electronic unit **130** of the first uniting unit UN**1** may be the same as the connection pattern **134** of the second electronic unit **130** of the second uniting unit UN**2** (e.g., the disconnection-preventing design DSNa shown in FIG. **3**), the connection pattern **134** of the second electronic unit **130** of the first uniting unit UN**1** may be different from the connection pattern **144** of the bridge unit **140** (e.g., the disconnection-preventing design DSNc shown in FIG. **3**) and the connection pattern **154** of the dummy unit **150** (e.g., the disconnection-preventing design DSNf shown in FIG. **3**), and the connection pattern **144** of the bridge unit **140** may be different from the connection pattern **154** of the dummy unit **150**, but not limited thereto.

As shown in FIG. **5** to FIG. **8**, for instance, the mesh pattern **132** and the connection pattern **134** of the second electronic unit **130** may be a single-layer conductive layer (e.g., a conductive layer CL**5**), the mesh pattern **132** of the bridge unit **140** may include a plurality of conductive layers (e.g., a conductive layer CL**4** and a conductive layer CL**5**), and the connection pattern **144** of the bridge unit **140** may be a single-layer conductive layer or include a plurality of conductive layers, but not limited thereto. In some embodiments (as shown in FIG. **6**), some portions of the mesh pattern **142** of the bridge unit **140** may have one conductive layer (e.g., the conductive layer CL**4** or the conductive layer CL**5**), other portion (s) of the mesh pattern **142** of the bridge unit **140** may include a plurality of conductive layers (e.g., the conductive layer CL**4** and the conductive layer CL**5**), and these portions may complement each other in the top view, so as to achieve a visual effect of the mesh pattern **132** in the top view, but not limited thereto. In some embodiments, the whole mesh pattern **142** of the bridge unit **140** may include a plurality of conductive layers (e.g., the conductive layer CL**4** and the conductive layer CL**5**), but not limited thereto.

The connection pattern **144** of the bridge unit **140** may be connected to the connection pattern **134** of the second electronic unit **130** of the first uniting unit UN**1** or the connection pattern **134** of the second electronic unit **130** of the second uniting unit UN**2** based on requirement(s). As shown in FIG. **5** to FIG. **8**, the connection pattern **134** (e.g., the conductive layer CL**5**) of the second electronic unit **130** of the first uniting unit UN**1** may be connected to the connection pattern **144** (e.g., the conductive layer CL**4**) of the bridge unit **140** through a hole passing through an insulating layer IL**5**, and the connection pattern **134** (e.g., the conductive layer CL**5**) of the second electronic unit **130** of the second uniting unit UN**2** may be directly connected to the connection pattern **144** (e.g., the conductive layer CL**5**) of the bridge unit **140**; that is to say, a structure of the bridge unit **140** connected to the first uniting unit UN**1** may be disposed in the conductive layer CL**4**, and another structure of the bridge unit **140** connected to the second uniting unit UN**2** may be disposed in the conductive layer CL**5**, but not limited thereto.

In FIG. **5** and FIG. **9**, the connection pattern **154** of the mesh pattern **152** of the dummy unit **150** may be electrically insulated from the second electronic unit **130** of the first uniting unit UN**1**, the second electronic unit **130** of the second uniting unit UN**2** and the bridge unit **140**. In some embodiments, the dummy unit **150** may not have the connection pattern **154** so as not to be connected to the second electronic unit **130** and the bridge unit **140**.

In the present disclosure, the electronic device **100** may optionally include other suitable component and/or structure. In FIG. **6** to FIG. **9**, the electronic device **100** may further include at least one conductive layer (e.g., a conductive layer CL**1**, a conductive layer CL**2** and a conductive layer CL**3**), at least one insulating layer (e.g., an insulating layer IL**1**, an insulating layer IL**2**, an insulating layer IL**3**, an insulating layer IL**4**, an insulating layer IL**5**) and at least one semiconductor layer, so as to form the needed component(s) and/or structure(s), such as a transistor SW, a scan line and a data line. The transistor SW may be electrically connected to the first electronic unit **120** (the light emitting component LE), the scan line and the data line may be connected between the electronic component in the active region and the electronic component in the peripheral region, the scan line may be configured to transmit switching signal(s) for turning on or turning off the transistor SW, and the data line may be configured to transmit the gray level signal(s) which the light emitting component LE needs, but not limited thereto. In some embodiments (as shown in FIG. **6**), the transistor SW may include a semiconductor layer SM, the insulating layer IL**2**, a gate (e.g., the gate may be included in the conductive layer CL**1**), a source (e.g., the source may be included in the conductive layer CL**2**) and a drain (e.g., the drain may be included in the conductive layer CL**2**), the scan line may be included in the conductive layer CL**1**, and the data line may be included in the conductive layer CL**2**, but not limited thereto. Note that, the material of the semiconductor layer SM may include such as poly-silicon, amorphous silicon, metal-oxide semiconductor (e.g., IGZO), other suitable semiconductor materials or a combination thereof, but not limited thereto.

The electronic device **100** may further include a pixel defining layer PDL disposed between two light emitting components LE or two sub-pixels SP, so as to separate the light emitting components LE or the sub-pixels SP. The pixel defining layer PDL may be a single-layer structure or a composite structure, and may include an insulating material (organic insulating material or inorganic insulating material), a reflective material, other suitable material(s) or a combination thereof, but not limited thereto.

The electronic device **100** may optionally include an encapsulating layer EC, so as to encapsulate and protect the first electronic unit **120** (the light emitting component LE), wherein the encapsulating layer EC may include organic insulating material or other suitable encapsulating material. In FIG. **6**, the second electronic unit **130** may be disposed on the encapsulating layer EC, but not limited thereto.

In FIG. **6** to FIG. **9**, the electronic device **100** may optionally include an elastic material layer ELS disposed on the first electronic unit **120** and the second electronic unit **130**. The elastic material layer ELS may also be disposed in the region where the stretchable substrate **110** does not exist (i.e., some portions of the elastic material layer ELS may be respectively disposed between two island portions **112**, between two bridge portions **114** and between two substrate units **110*u***), so as to improve the structural strength and reliability of the electronic device **100**. The elastic material layer ELS may include such as transparent elastic insulating material, but not limited thereto.

The electronic device of the present disclosure is not limited to the above embodiments. Further embodiments of the present disclosure are described below. For ease of comparison, same components will be labeled with the same symbol in the following. The following descriptions relate the differences between each of the embodiments, and repeated parts will not be redundantly described.

Referring to FIG. **10**, FIG. **10** is a schematic diagram showing a cross-sectional view of an electronic device according to a second embodiment of the present disclosure. As shown in FIG. **10**, a difference between this embodiment and the first embodiment is the position of the second electronic unit **130** of the electronic device **200**. In FIG. **10**, the second electronic unit **130** may be disposed between the first electronic unit **120** (the light emitting component LE) and the transistor SW, but not limited thereto.

Referring to FIG. **11**, FIG. **11** is a schematic diagram showing a cross-sectional view of an electronic device according to a third embodiment of the present disclosure. As shown in FIG. **11**, a difference between this embodiment and the first embodiment is the position of the second electronic unit **130** of the electronic device **300**. In FIG. **11**, the second electronic unit **130** may be disposed between the encapsulating layer EC and the pixel defining layer PDL, but not limited thereto. Moreover, in the connection pattern **134** on the bridge portion **114** of the stretchable substrate **110**, the connection pattern **134** may be a multi-layer structure including the conductive layer CL**5** and the conductive layer CL**2**. In the connection pattern **134**, a portion included in the conductive layer CL**5** may be electrically connected to another portion included in the conductive layer CL**2** through a hole passing through the insulating layer IL**4**, such that the conductive layer CL**2** may include the source and the drain of the transistor SW and the another portion of the connection pattern **134** of the second electronic unit **130**. In some embodiments, the connection pattern **134** may horizontally extend through the another portion included in the conductive layer CL**2**, but not limited thereto. In some embodiments, if the connection pattern **144** of the bridge unit **140** has this design, the number of conductive layers used in the electronic device **300** may be reduced, thereby reducing the cost of the electronic device **300**.

Referring to FIG. **12**, FIG. **12** is a schematic diagram showing a cross-sectional view of an electronic device according to a fourth embodiment of the present disclosure. As shown in FIG. **12**, a difference between this embodiment and the first embodiment is the design of the connection pattern **134** of the second electronic unit **130** of the electronic device **400**. In FIG. **12**, in the connection pattern **134** on the bridge portion **114** of the stretchable substrate **110**, the connection pattern **134** may be a multi-layer structure including the conductive layer CL**4** and the conductive layer CL**5**. In the connection pattern **134**, a portion included in the conductive layer CL**5** may be electrically connected to another portion included in the conductive layer CL**4** through a hole passing through the insulating layer. In some embodiments, the connection pattern **134** may horizontally extend through the another portion included in the conductive layer CL**4**, but not limited thereto.

Referring to FIG. **13**, FIG. **13** is a schematic diagram showing a cross-sectional view of an electronic device according to a fifth embodiment of the present disclosure. As shown in FIG. **13**, differences between this embodiment and the first embodiment are the design of the stretchable substrate **110** of the electronic device **500** and the configurations of the first electronic unit **120** and the second electronic unit **130** of the electronic device **500**. In FIG. **13**, the island portion **112** may be U-shaped, one island portion **112** may be directly connected to five bridge portions **114**, and each substrate unit **110*u*** may include two island portions **112** and nine bridge portions **114** directly connected to these island portions **112**, wherein one first bridge portion **114*a*** may be connected between these island portions **112**, and each of one ends of eight second bridge portions **114*b*** may be connected to the island portion **112** of other substrate unit **110*u***. In some embodiments (as shown in FIG. **13**), a width of the first bridge portion **114*a*** may be greater than a width of the second bridge portion **114*b***, but not limited thereto. In some embodiments (as shown in FIG. **13**), the stretchable substrate **110** may have a H-shaped hole **110*h***, and the top view pattern of two ends of the H-shaped hole **110*h*** may be similar to a triangle, but not limited thereto.

For example, as shown in FIG. **13**, one island portion **112** may have two pixels, and one pixel may include such as four sub-pixels SP, but not limited thereto. In one island portion **112** shown in FIG. **13**, two pixels are disposed on two sides of the first bridge portion **114*a*** in the top view, but not limited thereto.

As shown in FIG. **13**, in one second electronic unit **130**, the mesh pattern **132** may include at least one first mesh pattern **132_1** and at least one second mesh pattern **132_2**, the connection pattern **134** may include at least one first connection pattern **134_1** and at least one second connection pattern **134_2**, wherein the first mesh pattern(s) **132_1** and the first connection pattern(s) **134_1** may be electrically connected to each other to form a first sub-unit US**1**, and the second mesh pattern **132_2** and the second connection pattern **134_2** may be electrically connected to each other to form a second sub-unit US**2**. The first sub-unit US**1** and the second sub-unit US**2** may not be directly connected to each other, such that the first sub-unit US**1** and the second sub-unit US**2** may be electrically insulated from each other and/or may not conduct to each other. In FIG. **13**, in one second electronic unit **130**, the first connection pattern(s) **134_1** may be disposed on at least one second bridge portion **114*b***, the second connection pattern(s) **134_2** may be disposed on at least one second bridge portion **114*b***, and the first connection pattern(s) **134_1** and the second connection pattern(s) **134_2** may be disposed on the first bridge portion **114*a***, but not limited thereto. In FIG. **13**, in one second electronic unit **130**, the first connection pattern **134_1** on the first bridge portion **114*a*** may be electrically connected to the first mesh patterns **132_1** respectively disposed on two different island portions **112**, and the second connection pattern **134_2** on the first bridge portion **114*a*** may be electrically connected to the second mesh patterns **132_2** respectively disposed on two different island portions **112**, but not limited thereto. Moreover, the second electronic unit **130** may further include a bridge component **530** disposed at the connecting position of the first bridge portion **114*a*** and the island portion **112**, the bridge component **530** may be electrically connected to the first connection pattern **134_1** on the first bridge portion **114*a*** and the first mesh pattern **132_1** through a first conductive path, and the bridge component **530** may be electrically connected to the second connection pattern **134_2** on the first bridge portion **114*a*** and the second mesh pattern **132_2** through a second conductive path, wherein the first conductive path is different from the second conductive path (e.g., conductive structures respectively in the different conductive layers of the bridge component **530** may serve as the first conductive path and the second conductive path respectively), such that the first sub-unit US**1** does not conduct to the second sub-unit US**2**. In FIG. **13**, in order to distinguish the first sub-unit US**1**, the second sub-unit US**2** and the bridge component **530**, the first sub-unit US**1** is shown with the finest line, the second sub-unit US**2** is shown with the medium line, and the bridge component **530** is shown with the coarsest line, but the line widths of these structures of the present disclosure are not limited thereto. In some embodiments, the line width of the first sub-unit US**1**, the line width of the second sub-unit US**2** and the line width of the bridge component **530** may be the same. It can be known that, one second electronic unit **130** shown in FIG. **13** may include one first sub-unit US**1**, one second sub-unit US**2** and two bridge components **530**, the first sub-unit US**1** may include all first mesh patterns **132_1** and all first connection patterns **134_1** disposed on one substrate unit **110***u*, the second sub-unit US**2** may include all second mesh patterns **132_2** and all second connection patterns **134_2** disposed on one substrate unit **110***u*, and a portion of the first sub-unit US**1**, a portion of the second sub-unit US**2** and one bridge component **530** may be disposed on one island portion **112**.

In the present disclosure, the arrangements and the connections of the first sub-units US**1** and the second sub-units US**2** may be designed based on requirement(s). For example, the first sub-units US**1** arranged along the direction Y may be electrically connected to each other to form a sub-unit column, the sub-unit columns may be arranged along the direction X, the second sub-units US**2** arranged along the direction X may be electrically connected to each other to form a sub-unit row, and the sub-unit rows may be arranged along the direction Y, but not limited thereto.

In addition, in the mesh pattern **132** of the second electronic unit **130** shown in FIG. **13**, some portions of the mesh pattern **132** (e.g., a portion of the first mesh pattern **132_1** and a portion of the second mesh pattern **132_2**) may have the first opening **132***b* overlapping at least one first electronic unit **120**, other portions of the mesh pattern **132** (e.g., another portion of the first mesh pattern **132_1** and another portion of the second mesh pattern **132_2**) may have a fourth opening **132***d* not overlapping the first electronic unit **120**, but not limited thereto.

Referring to FIG. **14**, FIG. **14** is a schematic diagram showing a cross-sectional view of an electronic device according to a sixth embodiment of the present disclosure. For instance, the electronic device **600** shown in FIG. **14** may be disposed on the window of the vehicle, but the electronic device **600** of the present disclosure may be disposed on any suitable object BS based on requirement(s). In the present disclosure, the configurations the active region AR and the peripheral region PR may be designed based on requirement(s). The electronic components **610** (e.g., the first electronic unit **120** and the second electronic unit **130**) having the function corresponding to the active region AR may be disposed on the substrate units **110***u* (or the island portions **112**) of the active region AR, and the peripheral region PR may be disposed on at least one outer side of the active region AR. For instance, in FIG. **14**, the peripheral region PR may surround the active region AR, but not limited thereto. Note that, if the peripheral region PR includes the substrate units **110***u* (or the island portions **112**) of the stretchable substrate **110**, the electronic component **610** (e.g., the first electronic unit **120** and the second electronic unit **130**) having the function corresponding to the active region AR may not be disposed on these substrate units **110***u* (or the island portions **112**).

In some embodiments, the peripheral region PR may include a peripheral circuit region PCR. For instance, in FIG. **13**, the peripheral circuit region PCR may be disposed on a lower side of the active region AR, but not limited thereto. In some embodiments, a driving component configured to drive the electronic component **610** in the active region AR may be disposed in the peripheral circuit region PCR. Optionally, connecting pads **640** may be disposed in the peripheral circuit region PCR, and a circuit board **630** may be electrically connected to the connecting pads **640** through connecting components **620**, such that electronic component(s) on the circuit board **630** may be electrically to the driving component in the peripheral circuit region PCR, but not limited thereto.

In summary, because of the patterned design of the stretchable substrate and the design of the electronic units disposed on the stretchable substrate, the effect and the reliability of the electronic device would be enhanced when the electronic device is stretched and/or deformed, and the lifetime of the electronic device would be increased.

Although the embodiments and their advantages of the present disclosure have been described as above, it should be understood that any person having ordinary skill in the art can make changes, substitutions, and modifications without departing from the spirit and scope of the present disclosure. In addition, the protecting scope of the present disclosure is not limited to the processes, machines, manufactures, material compositions, devices, methods and steps in the specific embodiments described in the description. Any person having ordinary skill in the art can understand the current or future developed processes, machines, manufactures, material compositions, devices, methods and steps from the content of the present disclosure, and then, they can be used according to the present disclosure as long as the same functions can be implemented or the same results can be achieved in the embodiments described herein. Thus, the protecting scope of the present disclosure includes the above processes, machines, manufactures, material compositions, devices, methods and steps. Moreover, each claim constitutes an individual embodiment, and the protecting scope of the present disclosure also includes the combination of each claim and each embodiment. The protecting scope of the present disclosure shall be determined by the appended claims.

Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the disclosure. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

What is claimed is:

1. An electronic device, comprising:

a stretchable substrate having a plurality of island portions and a plurality of bridge portions, each of the bridge portions connecting at least two of the island portions;

a plurality of first electronic units disposed on the plurality of island portions; and

a plurality of second electronic units disposed on the stretchable substrate, each of the second electronic units comprising a mesh pattern disposed on at least one of the island portions, the mesh pattern comprising a first opening overlapped with at least one of the plurality of first electronic units;

wherein the plurality of second electronic units are sensing electrodes, and the mesh pattern is a portion of the sensing electrodes.

2. The electronic device according to claim **1**, wherein each of the second electronic units further comprises a connection pattern disposed on at least one of the bridge portions and the connection pattern comprises a second opening.

3. The electronic device according to claim **2**, wherein the first opening has a first area, the second opening has a second area and the first area is greater than the second area.

4. The electronic device according to claim **3**, wherein a ratio of the first area to the second area is greater than 50 and less than or equal to 5000.

5. The electronic device according to claim **3**, wherein the second opening is in a shape of circle, oval, rhombus, or bar.

6. The electronic device according to claim **1**, wherein the mesh pattern further comprises a third opening overlapped with at least another one of the plurality of first electronic units, the first opening has a first area, the third opening has a third area and a ratio of the first area to the third area is greater than 1 and less than or equal to 5.

7. The electronic device according to claim **1**, wherein each of the second electronic units further comprises a connection pattern disposed on at least one of the bridge portions and the connection pattern has a wavy shape.

8. The electronic device according to claim **1**, wherein one of the second electronic units is disposed on at least two of the island portions.

9. The electronic device according to claim **1**, further comprising a plurality of bridge units disposed on the stretchable substrate, wherein each of the bridge units electrically connects two adjacent ones of the second electronic units.

10. The electronic device according to claim **1**, further comprising a plurality of dummy units disposed on the stretchable substrate, wherein the plurality of dummy units are disposed between and electrically insulated from two adjacent ones of the second electronic units.

11. The electronic device according to claim **1**, wherein the plurality of first electronic units are light emitting diodes.

12. The electronic device according to claim **1**, wherein the plurality of first electronic units are photodiodes.

13. The electronic device according to claim **1**, wherein the plurality of second electronic units are touch electrodes.

14. The electronic device according to claim **1**, wherein the plurality of second electronic unit are configured to sense a stylus pen.

15. The electronic device according to claim **1**, wherein the plurality of second electronic units are configured to receive or transmit electromagnetic wave.

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